第六章 湍流概论

引言



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引言

目前难以给湍流下一个科学的定义。在工程技术中，称为“紊流”；而在日语中，湍流的汉字是“乱流”，语言上给出了一种形象的描述。

在流体力学中，类似的例子有“旋涡”。目前难以给“旋涡”下一个确切的科学定义。著名的空气动力学家 Dietrich Küchemann 将旋涡形象地比喻为“流体运动的肌腱”。

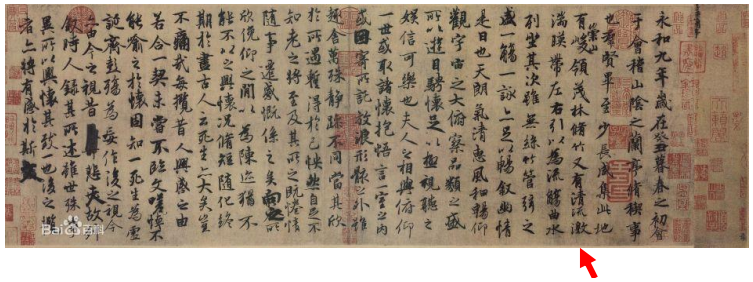
通过对烟囱排烟运动的观察，可以对湍流的性质有一个大概的了解，但很难对湍流下一个精确的定义，我们只能列出湍流流动的一些特性。

——H. Tennekes & J. L. Lumley

《A first course in turbulence》

研究历史

- 19世纪中叶以前
  - 公元363年，王羲之《兰亭序》：
    此地有崇山峻岭，茂林修竹；又有**清流激湍**
  - 1500，L. da Vinci，认识到流体运动的两种状态
  - 1839，G. Hagen，流体运动两种状态的再次发现



- 19世纪末—湍流理论研究的新纪元
  - 1883，O. Reynolds，圆管流动实验
    提出流动从层流向湍流转变的临界雷诺数的概念
  - 1887，Kelvin引入专用词“turbulence”
  - 1895，雷诺平均理论、雷诺方程、雷诺应力

- 20世纪初—湍流模式理论的雏形
  - 1877，Boussinesq，涡粘假设
  - 1925，Prandtl，混合长理论
  - 1930，von Karman，局部相似理论



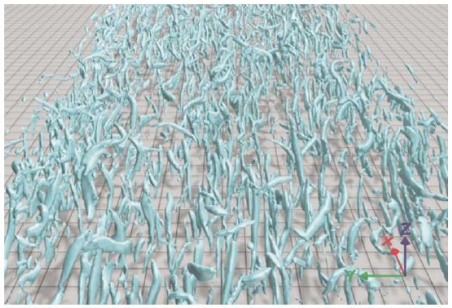
Figure 1.1 Leonardo da Vinci's turbulence model. (From Ritchter (1970).)

16世纪达芬奇描述的湍流

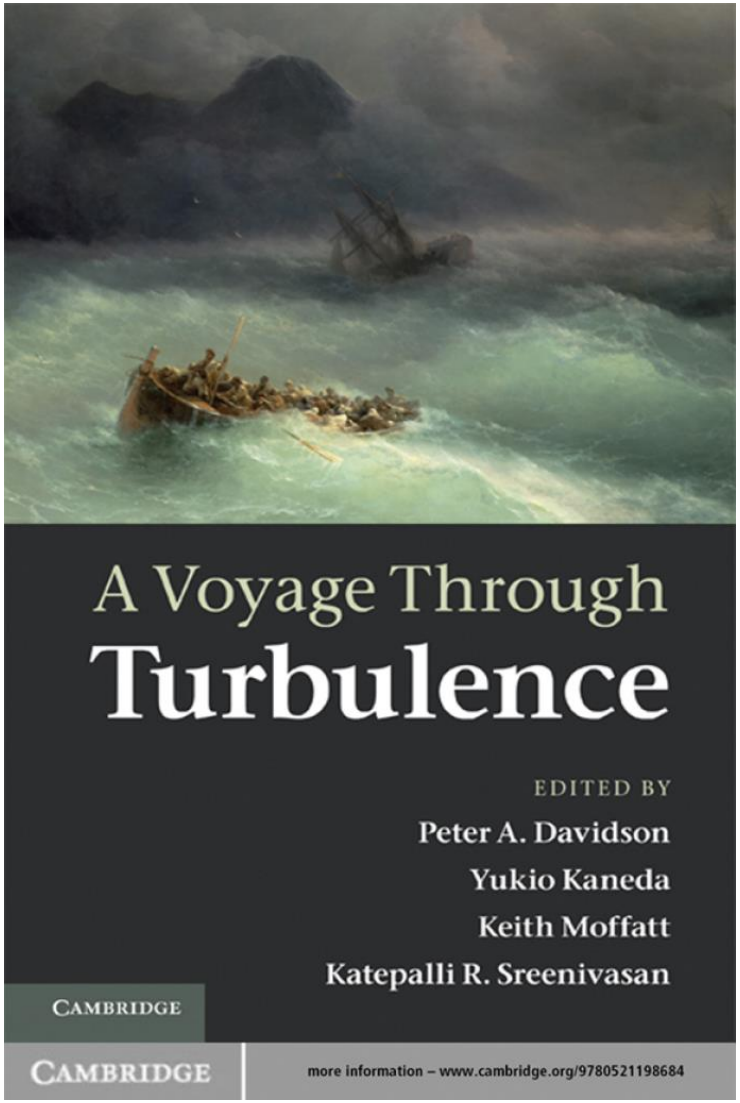
（另：南宋马远水图卷）

研究历史

- 20世纪30-40年代—湍流统计理论的发展
  - 1922，L.F. Richardson，能量传输的级串过程
  - 1930s，热线风速仪技术的发展，实现湍流脉动的测量
  - 1935，G.I. Taylor，均匀各向同性湍流
  - 1938，Karman-Howarth方程，
  - 1941，Kolmogrov，局部各向同性假设，-5/3次方能谱
- 20世纪40-70年代—湍流模式理论的蓬勃发展
  - 1940，P.Y. Zhou，雷诺应力输运方程，二阶矩模式理论
  - 1951，Rota，二阶矩模式的发展
  - 1970s，Launder, k-ε模型
- 20世纪70年代以后—湍流拟序结构研究的兴起
  - 1967，S.J. Kline，条带结构，猝发现象
  - 1976，A. Roshko，大尺度相干结构
  - 1980s，Adrian，PIV技术的发展
  - 1980s，J. Kim & P. Moin，湍流DNS的发展
  - 2000以后，I. Marusic，超大尺度相干结构？



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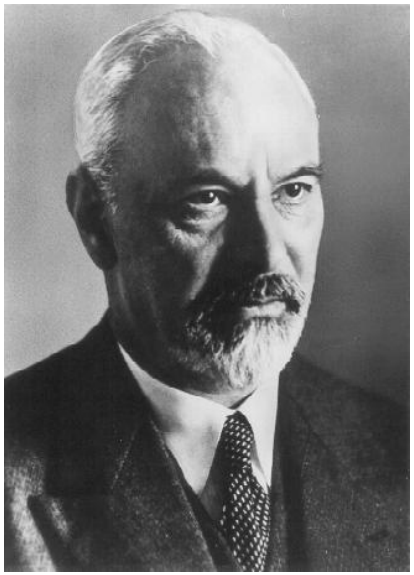
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研究历史

- 20世纪湍流研究的奠基人



L. Prandtl

(1875-1953)



G.I. Taylor

(1886-1975)



A. Kolmogrov

(1903-1987)



P.Y. Zhou

(1902-1993)

研究手段

- 量纲分析
- 渐进分析
- 统计分析
- 维象模型
- 实验观测
- 数值模拟

尝试解决湍流问题的成功与否，强烈地取决于包括做出关键性假设的灵感。

湍流需要奔放的发明者，正如需要专业分析师那么严重。

——H. Tennekes & J. L. Lumley

基本内容

§6.1　湍流的产生

§6.2　湍流的基本特征

§6.3　湍流运动的基本方程

§6.4　壁湍流的统计特性

§6.5　壁湍流相干结构

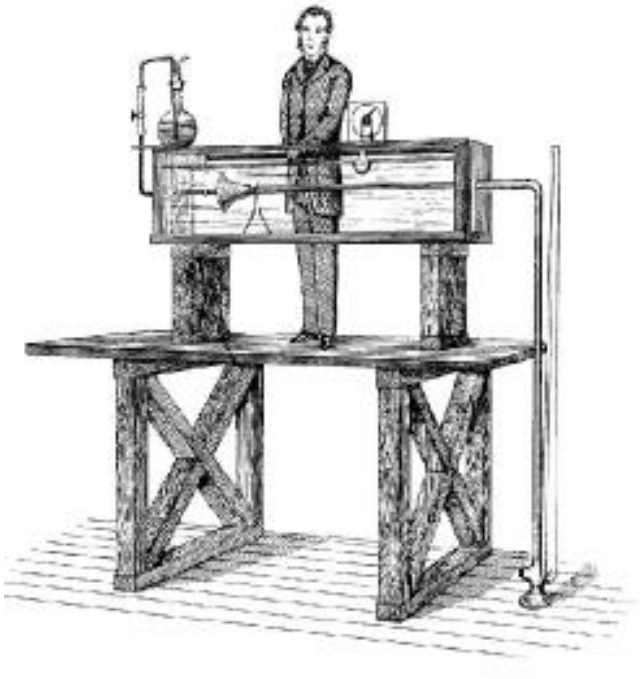
稳定性分析的重要性

For any problem of viscous flow under steady conditions there must in principle exist an exact steady solution of the equations of fluid dynamics. These solutions formally exist for all Reynolds numbers. Yet not every solution of the equations of motion, even if it is exact, can actually occur in Nature. Those which do must not only obey the equations of fluid dynamics, but also be stable. Any small perturbations which arise must decrease in the course of time. If, on the contrary, the small perturbations which inevitable occur in the flow tend to increase with time, the flow is unstable and cannot actually exist.

L. D. Landau & E.M. Lifshitz, Fluid Dynamics, Second Edition, 1959, P95

§6.1 湍流的产生

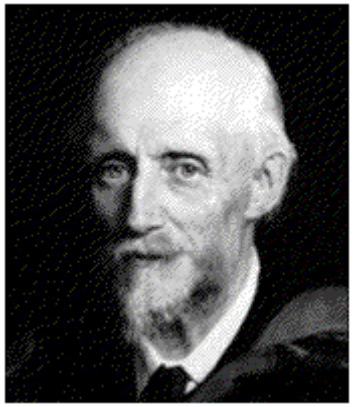
一、雷诺圆管流动显示实验



雷诺1883年实验装置

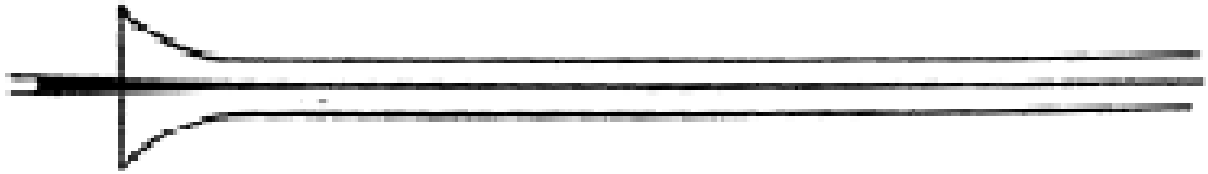


英国曼彻斯特大学



Osborne Reynolds
（1842－1912）

§6.1 湍流的产生



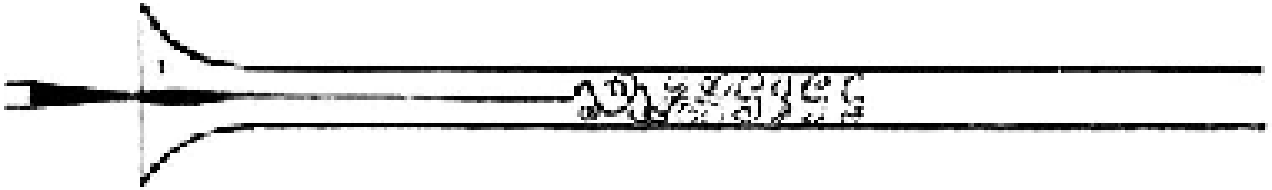
层流

Laminar Flow



湍流

Turbulent Flow



湍流中的涡结构

Eddy Structure

§6.1 湍流的产生

流体的运动存在两种截然不同的状态：层流和湍流

在某些条件下，流动可以从层流转变为湍流
从层流向湍流的过渡称为转捩（Transition）

控制流动状态的参数为雷诺数 $Re = U_m D / \nu$

雷诺数存在上下两个临界值:

下临界雷诺数 Re_{c1}：若Re< Re_{c1}流动为层流

上临界雷诺数 Re_{c2}：若$Re>$ Re_{c2}流动为**湍流**

雷诺测得： Re_{c1} = 2020　　Re_{c2} = 12830

近代实验：1760< Re_{c1}<2300　　Re_{c2}~10^5

§6.1 湍流的产生

二、流动稳定性和转捩

转捩与扰动大小有关

⟹ 转捩是扰动放大导致流动失稳的过程

流动稳定性问题是指流动对作用于它的小扰动如何响应的问题：

- 如果流动使该扰动逐渐减弱且最后消失，使整个流动能恢复到受扰动作用以前的状态，则流动是稳定的；
- 如果该扰动逐渐增长，或始终以某种方式存在，使流动不能恢复到受扰动作用以前的状态，而转变为另一种状态（可以是湍流也可以是另一种层流），则流动是不稳定的。

§6.1 湍流的产生

- 在雷诺实验的例子中，

  $Re < Re_{c1}$ 时，流动在扰动作用下仍能保持层流状态，表明流动是稳定的；

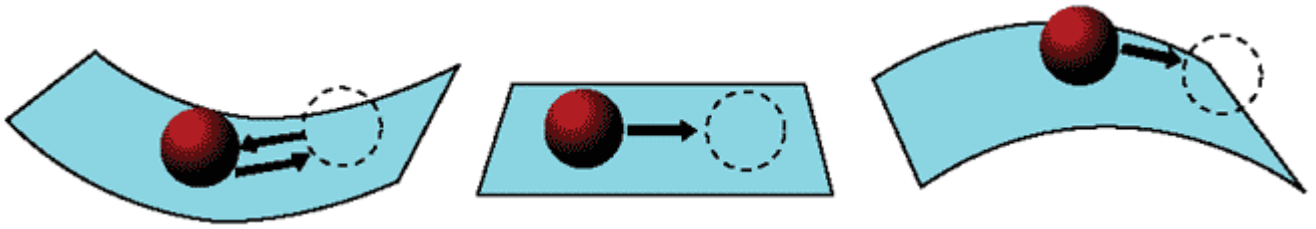
  $Re > Re_{c1}$ 时，在某些扰动下，流动可以从层流变为湍流状态，表明这个雷诺数下的层流流动是不稳定的 。

- 物理上存在的层流解不但满足动力学方程组，还应满足稳定性条件
- 流动失稳是转捩的开始，而不是转捩的全过程

§6.1 湍流的产生

三、流动不稳定机理和典型流动稳定性问题

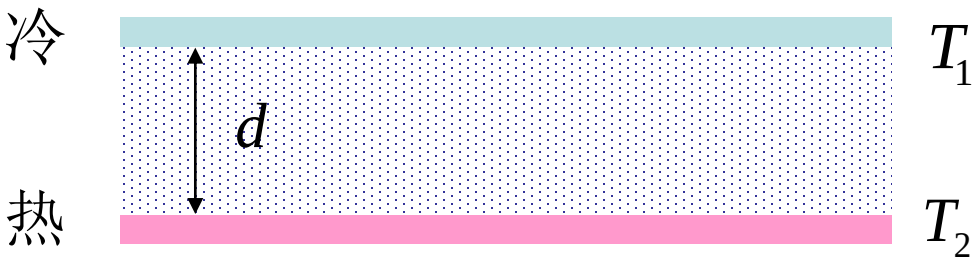
- 流动不稳定性是由于外力、惯性力和粘性力之间的平衡受到干扰而产生的
- 外力：浮力，表面张力，离心力，科氏力等
- 例如
  - Benard涡：浮力
  - Taylor涡：离心力
  - 雷诺圆管实验：惯性力和粘性力之间的平衡关系



Stable **Neutral** **Unstable**

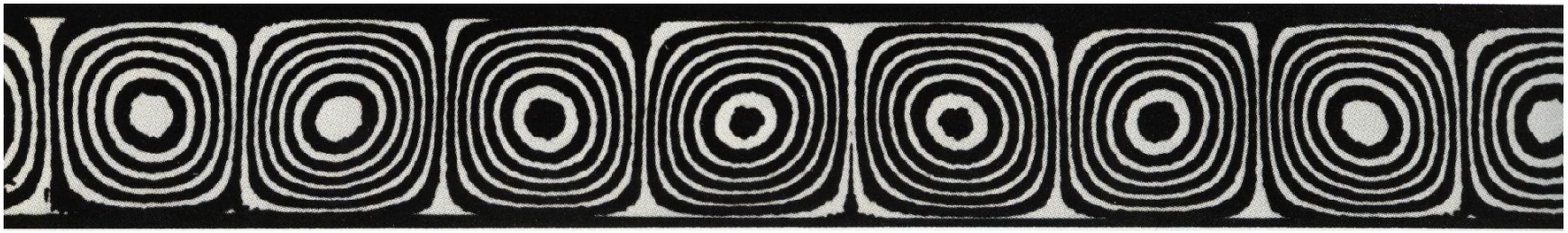
§6.1 湍流的产生

Benard 对流



Rayleigh数

$$Ra \equiv \frac{\gamma g \left(T_2 - T_1\right) d^3}{\nu \kappa}$$

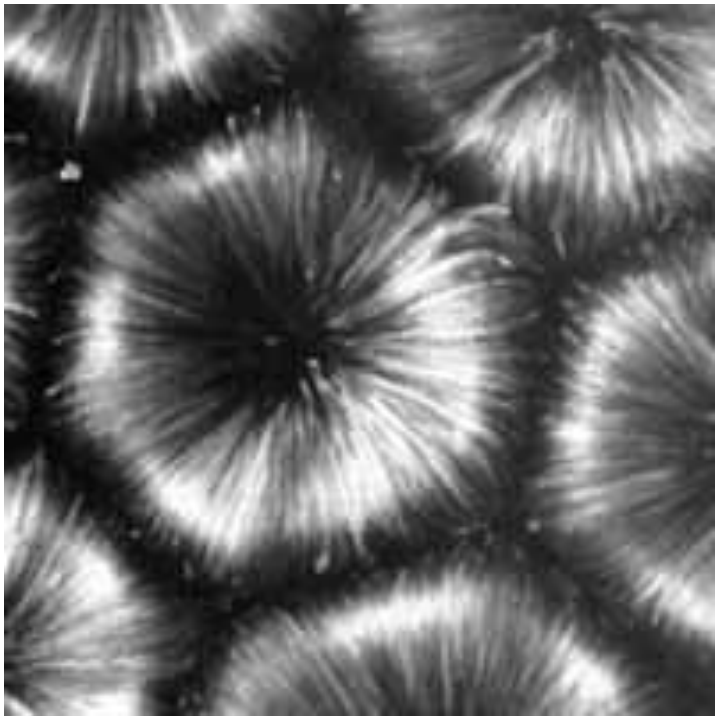


两水平平板之间形成的Benard涡。
两平板之间充满硅油，平板的长、宽和平板间距之比为10:4:1
下板加热，形成的涡平行于平板短边。

§6.1 湍流的产生

- **Benard 对流**



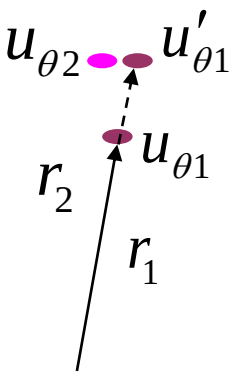


动画1, 2, 3, 4

§6.1 湍流的产生

离心不稳定性

Rayleigh准则：
若环量的平方随半径增大而增加，则流动稳定
若环量的平方随半径增大而降低，则流动不稳定



$$\frac{u_\theta^2}{r} = -\frac{1}{\rho}\frac{\partial p}{\partial r}$$ 离心力与压力梯度平衡

质点由位置1跑到位置2

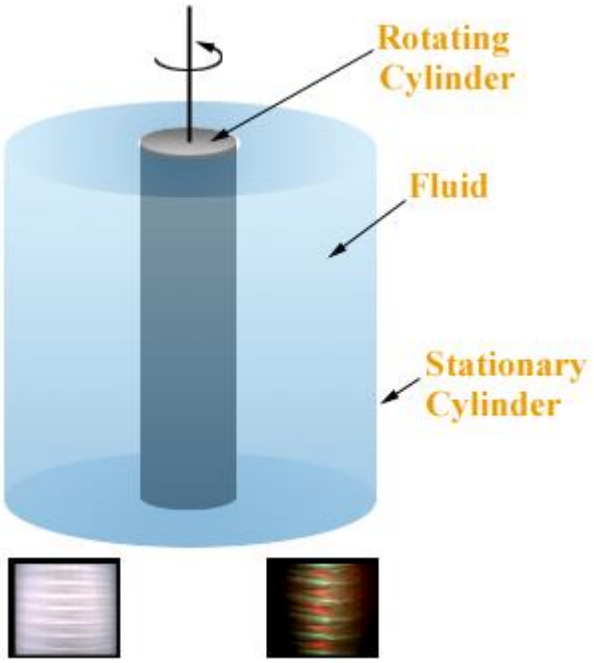
速度 $u'_{\theta 1} = \dfrac{r_1 u_{\theta 1}}{r_2}$ 离心力 $\dfrac{\left(u'_{\theta 1}\right)^2}{r_2} = \dfrac{\left(r_1 u_{\theta 1}\right)^2}{r_2^3}$

若位置2的压力梯度大于质点的离心力，则质点回到位置1：稳定

$$\left(-\frac{1}{\rho}\frac{\partial p}{\partial r}\right)_2 = \frac{u_{\theta 2}^2}{r_2} > \frac{\left(r_1 u_{\theta 1}\right)^2}{r_2^3} \Rightarrow \left(r_2 u_{\theta 2}\right)^2 > \left(r_1 u_{\theta 1}\right)^2 \Rightarrow \frac{\partial\left(r u_\theta\right)^2}{\partial r} > 0$$

§6.1 湍流的产生

离心不稳定性



动画　　动画

两同轴旋转圆筒间形成的Taylor涡。外圆柱不动，内圆柱旋转

Taylor 数

$$Ta \equiv -\frac{4A\omega_0 d^4}{\nu^2}$$

$$A \equiv \frac{\omega_1 R_1^2 - \omega_2 R_2^2}{R_1^2 - R_2^2}$$

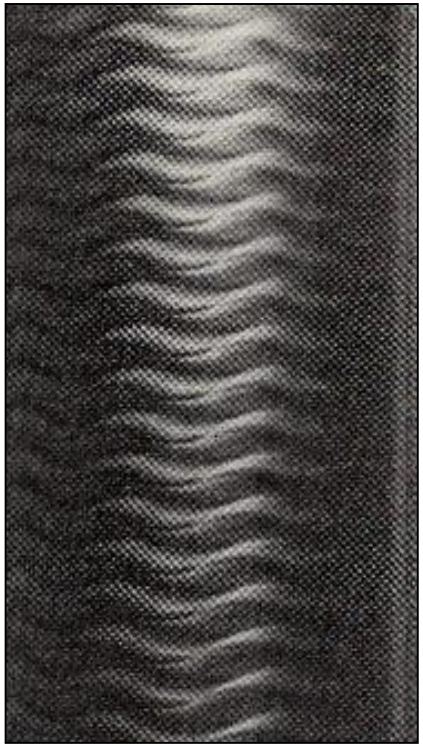
$$\omega_0 \equiv \frac{\omega_1 + \omega_2}{2}$$

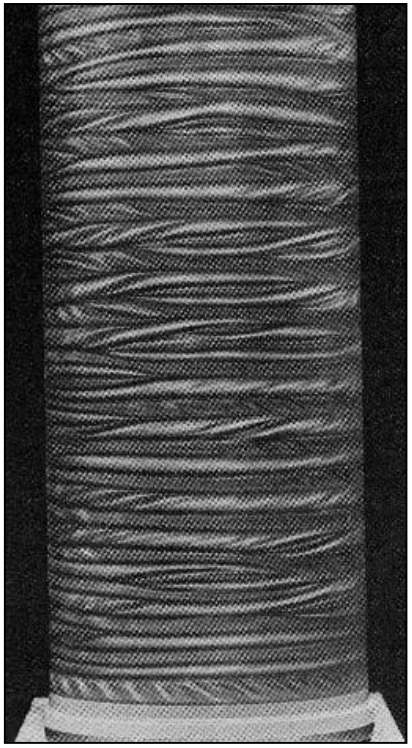
$$d \equiv R_2 - R_1$$

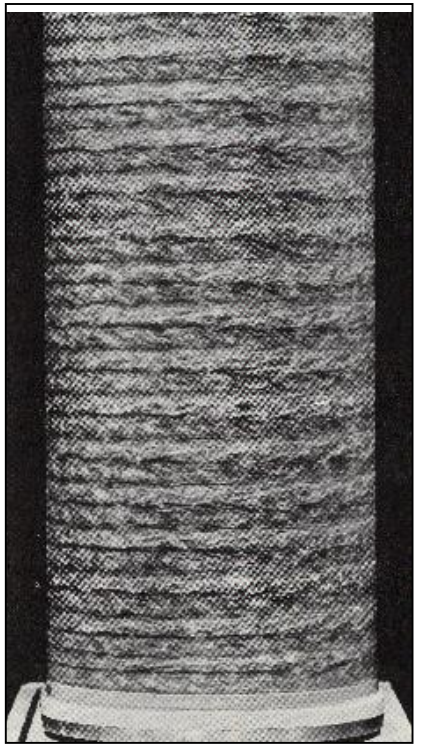
§6.1 湍流的产生

离心不稳定性

When the Reynolds number increases, the flow within the annulus transforms from a circular Couette flow → Taylor vortex flow → wavy vortex flow → modulated wavy vortex flow → turbulent Taylor vortex flow.







四、流动稳定性问题的建立

不可压缩牛顿型流体的运动满足

$$\frac{\partial u_i}{\partial t}+u_j\frac{\partial u_i}{\partial x_j}=-\frac{1}{\rho}\frac{\partial p}{\partial x_i}+\nu\frac{\partial^2 u_i}{\partial x_j\partial x_j}$$

$$\frac{\partial u_i}{\partial x_i}=0$$

$t=0$ 时： $u_i(\vec{x},0)=u_i^0(\vec{x})$

在Σ上： $u_i(\vec{x},t)|_\Sigma=u_i^\Sigma(\vec{x},t)$

在某点 A 上: $p(\vec{x}_A,t)=p_A$

流动稳定性

↓

N-S 方程初边值问题的解的稳定性

满足以上方程和初边值条件的一个解 (u_i,p)称为一个流动

§6.1 湍流的产生

流动稳定性的定义和临界雷诺数

初值稳定性问题的数学提法：

设一个流动(u_i, p)，当对其初值 u_i^0 施加扰动 $u_i^0{}'$ 后，形成一新的流动 $(\tilde{u}_i, \tilde{p})$，如果新的流动 $(\tilde{u}_i, \tilde{p})$ 以某种收敛的定义收敛到 (u_i, p)，

$$\lim_{t>t_0} \mathbb{F}(u_i - \tilde{u}_i) = 0 \qquad \lim_{t>t_0} \mathbb{F}(p - \tilde{p}) = 0$$

则称该流动是稳定的，否则是不稳定的。

定义扰动 $u_i' = \tilde{u}_i - u_i$ 扰动能量 $E_\forall = \iiint_\forall \frac{1}{2}\rho u_i' u_i' d\forall$

若 $\lim_{t\to\infty} \frac{E_\forall(t)}{E_\forall(0)} \to 0$，则称流动稳定

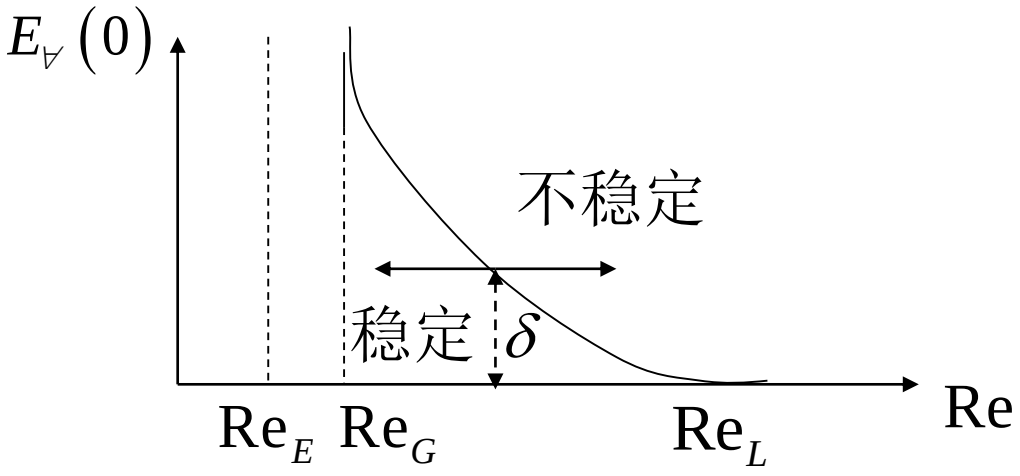
§6.1 湍流的产生

单调稳定 $\dfrac{\mathrm{d}E_{\forall}(t)}{\mathrm{d}t}<0,\quad \forall t \qquad \mathrm{Re}<\mathrm{Re}_E$

条件稳定 $E_{\forall}(0)<\delta$

普遍稳定 $\delta\to\infty \qquad \mathrm{Re}<\mathrm{Re}_G$ 可维持湍流的最低雷诺数

线性稳定 $\delta\to 0 \qquad \mathrm{Re}<\mathrm{Re}_L$ 可保持层流的最高雷诺数



§6.1 湍流的产生

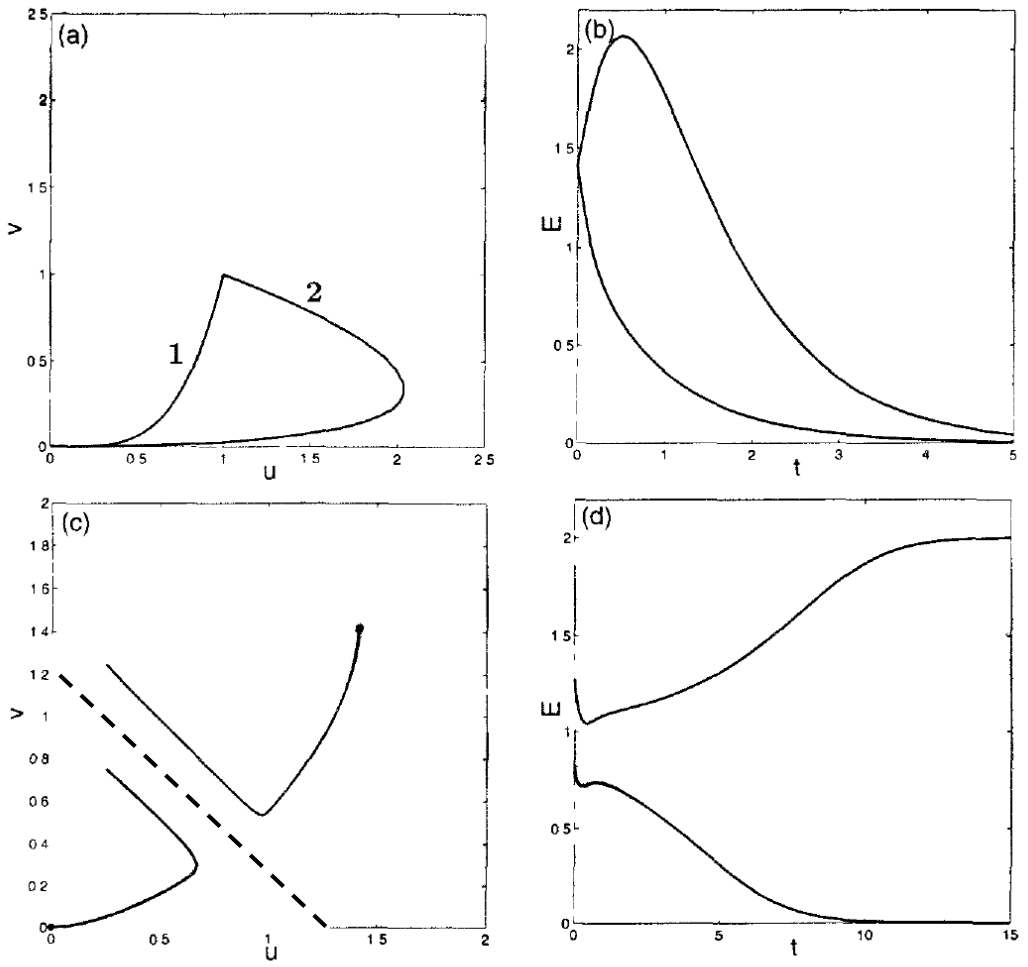


FIGURE 1 1 Sketch demonstrating different stability definitions (a,b) Monotonic versus non-monotonic stability The fixed point (0.0) is stable (c,d) Conditional stability The fixed point (0,0) is conditionally stable For initial conditions outside the gray triangular area the fixed point (0.0) is unstable

§6.1 湍流的产生

流动类型	Re_E	Re_G	Re_L	Re_T
Hagen-Poiseuille	81.5	1251	∞	2000
Plane-Poiseuille	49.6	—	5772	1000
Plane-Couette	20.7	125	∞	360
Blasius	—	—	515	—

§6.1 湍流的产生

扰动方程

基本流 (u_i, p)

$$\frac{\partial u_i}{\partial t}+u_j\frac{\partial u_i}{\partial x_j}=-\frac{1}{\rho}\frac{\partial p}{\partial x_i}+\nu\frac{\partial^2 u_i}{\partial x_j\partial x_j}$$

$$\frac{\partial u_i}{\partial x_i}=0$$

$t=0$ 时： $u_i(\vec{x},0)=u_i^0(\vec{x})$

在Σ上： $u_i(\vec{x},t)|_\Sigma=u_i^\Sigma(\vec{x},t)$

在某点A上： $p(\vec{x}_A,t)=p_A$

受扰流动 $(\tilde{u}_i, \tilde{p})$

$$\frac{\partial \tilde{u}_i}{\partial t}+\tilde{u}_j\frac{\partial \tilde{u}_i}{\partial x_j}=-\frac{1}{\rho}\frac{\partial \tilde{p}}{\partial x_i}+\nu\frac{\partial^2 \tilde{u}_i}{\partial x_j\partial x_j}$$

$$\frac{\partial \tilde{u}_i}{\partial x_i}=0$$

$\tilde{u}_i(\vec{x},0)=u_i^0(\vec{x})+u_i^{0\prime}(\vec{x})$

$\tilde{u}_i(\vec{x},t)|_\Sigma=u_i^\Sigma(\vec{x},t)$

$\tilde{p}(\vec{x}_A,t)=p_A$

§6.1 湍流的产生

定义：受扰流动$(\tilde{u}_i, \tilde{p})$与基本流(u_i, p)之差(u_i', p')为扰动

$$u_i' = \tilde{u}_i - u_i \qquad p' = \tilde{p} - p$$

扰动满足的方程和初边值条件为

$$\begin{cases} \dfrac{\partial u_i'}{\partial t} + u_j \dfrac{\partial u_i'}{\partial x_j} + u_j' \dfrac{\partial u_i}{\partial x_j} + u_j' \dfrac{\partial u_i'}{\partial x_j} = -\dfrac{1}{\rho}\dfrac{\partial p'}{\partial x_i} + \nu \dfrac{\partial^2 u_i'}{\partial x_j \partial x_j} \\ \dfrac{\partial u_i'}{\partial x_i} = 0 \\ u_i'|_{t=0} = u_i^{0'} \\ u_i'|_{\Sigma} = 0 \\ p'|_A = 0 \end{cases}$$

(u_i', p')的演化依赖于

基本流 u_i

扰动初值 $u_i^{0'}$

边界形状 Σ

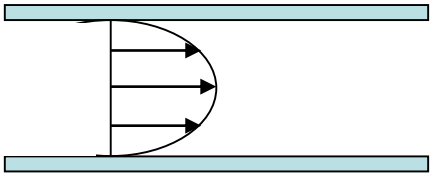
流体粘度 ν

在给定 $u_i, u_i^{0'}, \Sigma, \nu$ 时，若 $\lim\limits_{t\to\infty} \mathcal{F}(u_i') = 0$，流动是稳定的，否则是不稳定的。

§6.1 湍流的产生

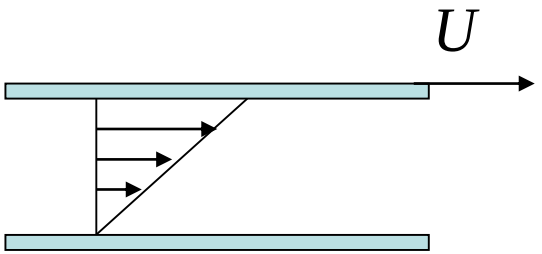
定常平面平行流：

$$u = u(y), \quad v = 0, \quad w = 0$$



平面Poiseuile流

$$u = u_c(1 - y^2), \quad v = 0, \quad w = 0$$



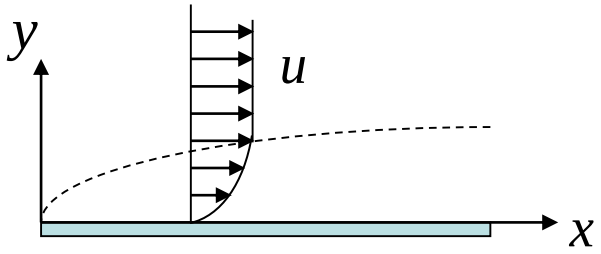
平面Couette流

$$u = Uy, \quad v = 0, \quad w = 0$$

边界层、射流、混合层等流动可近似为平面平行流

$$u = u(\varepsilon x, y), \quad v \sim \varepsilon u, \quad w \sim \varepsilon u$$

$$\varepsilon \to 0$$



§6.1 湍流的产生

粘性流动的线性扰动方程

线化扰动方程

$$\begin{cases} \dfrac{\partial u_i'}{\partial t}+u_j\dfrac{\partial u_i'}{\partial x_j}+u_j'\dfrac{\partial u_i}{\partial x_j}=-\dfrac{1}{\rho}\dfrac{\partial p'}{\partial x_i}+\nu\dfrac{\partial^2 u_i'}{\partial x_j\partial x_j} \\ \dfrac{\partial u_i'}{\partial x_i}=0 \end{cases}$$

定常平面平行流

$$\begin{cases} \dfrac{\partial u'}{\partial t}+u\dfrac{\partial u'}{\partial x}+v'\dfrac{\mathrm{d}u}{\mathrm{d}y}=-\dfrac{1}{\rho}\dfrac{\partial p'}{\partial x}+\nu\left(\dfrac{\partial^2 u'}{\partial x^2}+\dfrac{\partial^2 u'}{\partial y^2}+\dfrac{\partial^2 u'}{\partial z^2}\right) \\ \dfrac{\partial v'}{\partial t}+u\dfrac{\partial v'}{\partial x}=-\dfrac{1}{\rho}\dfrac{\partial p'}{\partial y}+\nu\left(\dfrac{\partial^2 v'}{\partial x^2}+\dfrac{\partial^2 v'}{\partial y^2}+\dfrac{\partial^2 v'}{\partial z^2}\right) \\ \dfrac{\partial w'}{\partial t}+u\dfrac{\partial w'}{\partial x}=-\dfrac{1}{\rho}\dfrac{\partial p'}{\partial z}+\nu\left(\dfrac{\partial^2 w'}{\partial x^2}+\dfrac{\partial^2 w'}{\partial y^2}+\dfrac{\partial^2 w'}{\partial z^2}\right) \\ \dfrac{\partial u'}{\partial x}+\dfrac{\partial v'}{\partial y}+\dfrac{\partial w'}{\partial z}=0 \end{cases}$$

§ 6.1 湍流的产生

$$\frac{\partial}{\partial x}:\quad \frac{\partial u'}{\partial t}+u\frac{\partial u'}{\partial x}+v'\frac{\mathrm{d}u}{\mathrm{d}y}=-\frac{1}{\rho}\frac{\partial p'}{\partial x}+\nu\nabla^2 u'$$

$$\frac{\partial}{\partial z}:\quad \frac{\partial w'}{\partial t}+u\frac{\partial w'}{\partial x}=-\frac{1}{\rho}\frac{\partial p'}{\partial z}+\nu\nabla^2 w'$$

$$\frac{\partial}{\partial t}\left(\frac{\partial u'}{\partial x}+\frac{\partial w'}{\partial z}\right)+u\frac{\partial}{\partial x}\left(\frac{\partial u'}{\partial x}+\frac{\partial w'}{\partial z}\right)+\frac{\partial v'}{\partial x}\frac{\mathrm{d}u}{\mathrm{d}y}=-\frac{1}{\rho}\left(\frac{\partial^2 p'}{\partial x^2}+\frac{\partial^2 p'}{\partial z^2}\right)+\nu\nabla^2\left(\frac{\partial u'}{\partial x}+\frac{\partial w'}{\partial z}\right)$$

$$\frac{\partial u'}{\partial x}+\frac{\partial v'}{\partial y}+\frac{\partial w'}{\partial z}=0 \quad\Longrightarrow\quad \frac{\partial u'}{\partial x}+\frac{\partial w'}{\partial z}=-\frac{\partial v'}{\partial y}$$

$$\frac{\partial}{\partial y}:\quad \frac{\partial}{\partial t}\left(\frac{\partial v'}{\partial y}\right)+u\frac{\partial}{\partial x}\left(\frac{\partial v'}{\partial y}\right)-\frac{\partial v'}{\partial x}\frac{\mathrm{d}u}{\mathrm{d}y}=\frac{1}{\rho}\left(\frac{\partial^2 p'}{\partial x^2}+\frac{\partial^2 p'}{\partial z^2}\right)+\nu\nabla^2\left(\frac{\partial v'}{\partial y}\right)$$

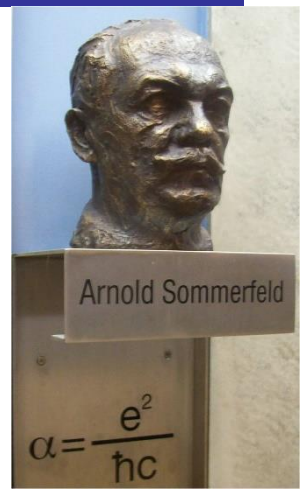
$$\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial z^2}:\quad \frac{\partial v'}{\partial t}+u\frac{\partial v'}{\partial x}=-\frac{1}{\rho}\frac{\partial p'}{\partial y}+\nu\nabla^2 v'$$

$$\frac{\partial}{\partial t}\left(\nabla^2 v'\right)+u\frac{\partial}{\partial x}\left(\nabla^2 v'\right)+\frac{\mathrm{d}u}{\mathrm{d}y}\frac{\partial}{\partial x}\left(\frac{\partial v'}{\partial y}\right)-\frac{\partial v'}{\partial x}\frac{\mathrm{d}^2 u}{\mathrm{d}y^2}-\frac{\partial^2 v'}{\partial x\partial y}\frac{\mathrm{d}u}{\mathrm{d}y}=\nu\nabla^2\left(\nabla^2 v'\right)$$

§6.1 湍流的产生

- **Orr-Sommerfeld方程**

$$\frac{\partial}{\partial t}\left(\nabla^2 v'\right)+u\frac{\partial}{\partial x}\left(\nabla^2 v'\right)-\frac{\mathrm{d}^2 u}{\mathrm{d}y^2}\frac{\partial v'}{\partial x}=\nu\nabla^2\left(\nabla^2 v'\right)$$



- **Squire方程**

$$\frac{\partial \omega_y'}{\partial t}+u\frac{\partial \omega_y'}{\partial x}+\frac{\mathrm{d}u}{\mathrm{d}y}\frac{\partial v'}{\partial z}=\nu\nabla^2\omega_y' \qquad \omega_y'=\frac{\partial u'}{\partial z}-\frac{\partial w'}{\partial x}$$

- **压力方程**

$$\nabla^2 p'=-2\frac{\mathrm{d}u}{\mathrm{d}y}\frac{\partial v'}{\partial x}$$

§6.1 湍流的产生

O-S方程的本征值问题

可分离变量形式的解 T-S波

$$\begin{Bmatrix} u' \\ v' \\ w' \\ p' \end{Bmatrix} = \begin{Bmatrix} \hat{u}(y) \\ \hat{v}(y) \\ \hat{w}(y) \\ \hat{p}(y) \end{Bmatrix} e^{-i\omega t} e^{i\alpha x + i\beta z}$$

ω 扰动频率

α 流向扰动波数

β 展向扰动波数

$$\left(D^2 - \alpha^2 - \beta^2\right)^2 \hat{v} - \frac{i\alpha}{\nu}\left(u - \frac{\omega}{\alpha}\right)\left(D^2 - \alpha^2 - \beta^2\right)\hat{v} + \frac{i\alpha}{\nu}\left(D^2 u\right)\hat{v} = 0$$

Orr-Sommerfeld方程： 四阶线性复变系数齐次方程

§6.1 湍流的产生

选取特征速度 U 和特征长度 L 对方程无量纲化

$$c \equiv \frac{\omega}{\alpha} = c_r + ic_i$$

$$\left(D^2-\alpha^2-\beta^2\right)^2\hat{v}-i\alpha Re(u-c)\left(D^2-\alpha^2-\beta^2\right)\hat{v}+i\alpha Re\left(D^2u\right)\hat{v}=0$$

O-S方程的边界条件：需要4个边界条件

（1）在固壁上： $\hat{v}=0 \quad D\hat{v}=0$

（2）在无穷远处 $y\to\infty$ ： $\hat{v}=0 \quad D\hat{v}=0$, 边界层，射流的外缘

（3）对称条件：$\hat{v}(0)=0 \quad D^2\hat{v}(0)=0$, u，w对称， v反对称

（4）反对称条件： $D\hat{v}(0)=0 \quad D^3\hat{v}(0)=0$，$u$，$w$反对称， v对称

注意：边界条件也是齐次的

§6.1 湍流的产生

- 本征值问题
  - 线性齐次方程满足齐次边界条件的非零解问题
  - 本征关系式：要有非零解，方程组系数必须满足的关系式 $\mathcal{F}\left(Re,\alpha,\beta,c\right)=0$
  - 本征解：非零解 $\hat{v}(y)$

§6.1 湍流的产生

- **本征值问题的一般解法**
  - 四阶线性方程有**4**个线性独立解 $\hat{v}_i, \quad i = 1,2,3,4$
  - 通解 $\hat{v} = \sum_{i=1}^{4} A_i \hat{v}_i$
  - 边界条件（以平面**Poiseuille**流为例）

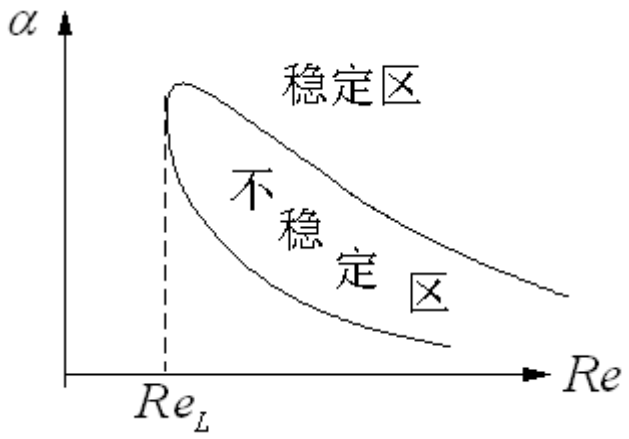
$$\hat{v}(1) = 0, \quad D\hat{v}(1) = 0, \quad \hat{v}(-1) = 0, \quad D\hat{v}(-1) = 0,$$

$$\begin{aligned}
&\sum_{i=1}^{4} A_i \hat{v}_i(1) = 0 \\
&\sum_{i=1}^{4} A_i D\hat{v}_i(1) = 0 \\
&\sum_{i=1}^{4} A_i \hat{v}_i(-1) = 0 \\
&\sum_{i=1}^{4} A_i D\hat{v}_i(-1) = 0
\end{aligned}
\quad \Rightarrow \quad
\begin{vmatrix}
\hat{v}_1(1) & \hat{v}_2(1) & \hat{v}_3(1) & \hat{v}_4(1) \\
D\hat{v}_1(1) & D\hat{v}_2(1) & D\hat{v}_3(1) & D\hat{v}_4(1) \\
\hat{v}_1(-1) & \hat{v}_2(-1) & \hat{v}_3(-1) & \hat{v}_4(-1) \\
D\hat{v}_1(-1) & D\hat{v}_2(-1) & D\hat{v}_3(-1) & D\hat{v}_4(-1)
\end{vmatrix} = 0$$

§6.1 湍流的产生

拇指曲线

$$\mathcal{F}\left(Re,\alpha,\beta,c\right)=0$$



对于时间模式问题，给定 Re,α,β

可确定本征值 $c=c_r+ic_i$

$c_i<0$ 流动是稳定的，

$c_i>0$ 流动是不稳定的，

$c_i=0$ 为中性情况。

给定 β 使得 $c_i=0$ 的 α 和 Re 在 $Re-\alpha$

平面上的分布曲线称为中性曲线

线性临界雷诺数

五、典型流动线性稳定性结果

$$\frac{1}{-i\alpha\,\mathrm{Re}}\left(D^2-\alpha^2\right)^2\hat{v}+u\left(D^2-\alpha^2\right)\hat{v}-\left(D^2u\right)\hat{v}=c\left(D^2-\alpha^2\right)\hat{v}$$

OS方程本征值问题的数值解法

1. 差分法

2.谱方法（正交函数展开法）

- **Chebyshev-Tau方法**
- **配置点法**

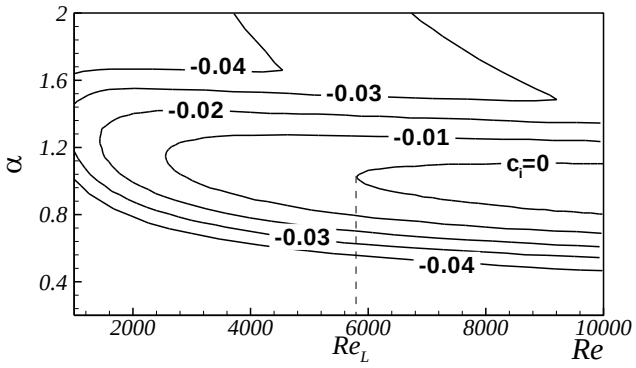
D.R.Gardner, S.A.Trogdon and R.W.Douglass, 1989. “A Modified Tau Spectral Method That Eliminates Spurious Eigenvalues”, Journal of Computational Physics,80: 137-167

G.B.McFadden, B.T.Murray and R.F.Boisvert,1990. “Elimination of Spurious Eigenvalues in the Chebyshev Tau Spectral Method”, Journal of Computational Physics, 91, 228-239

W-Z.Huang and D.M.Sloan, 1994. “The Pseudospectral Method for solving Differential Eigenvalue Problems”, Journal of Computational Physics, 111, 399-409

§6.1 湍流的产生

– **平面 Poiseuille流**

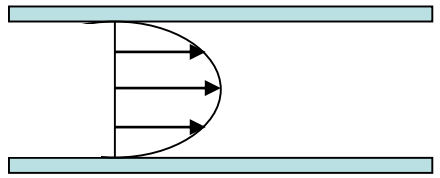


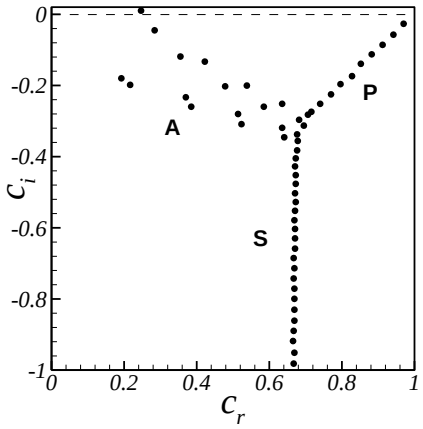
c_i 的等值线

$Re_L = 5772.2$

$\alpha = 1.020$

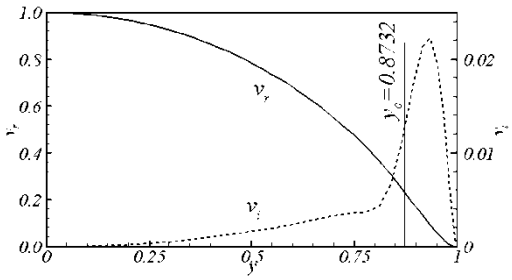
$c_r = 0.2639$

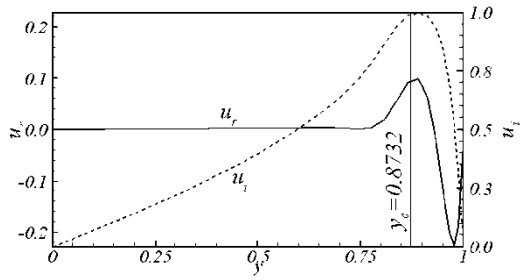




O-S方程的本征值和本征模态 $Re = 10^4$ $\alpha = 1$

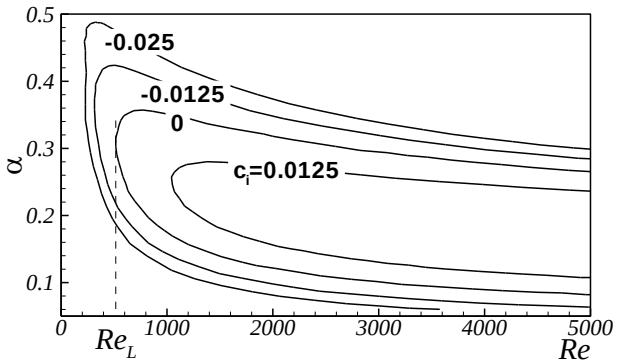
$c_i = 0.00373967 > 0$ 最不稳定模态





§6.1 湍流的产生

– **Blasius边界层**

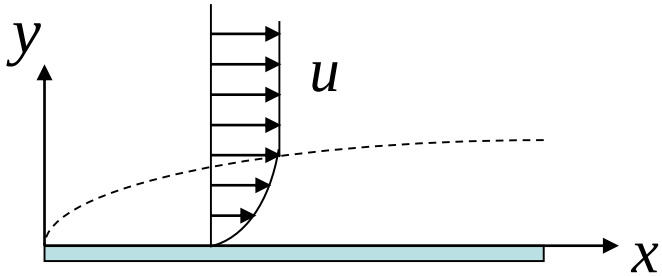


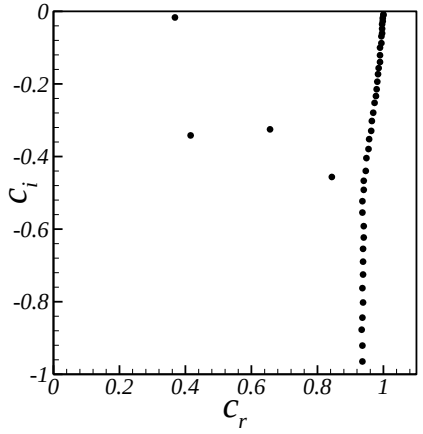
c_i 的等值线

$Re_L = 519.4$

$\alpha = 0.303$

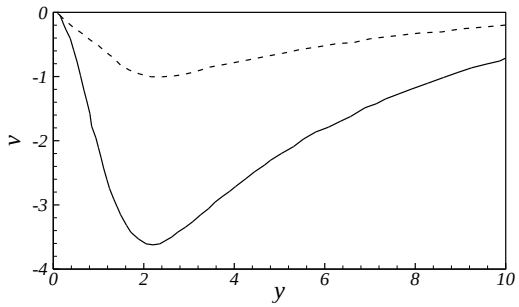
$c_r = 0.3965$

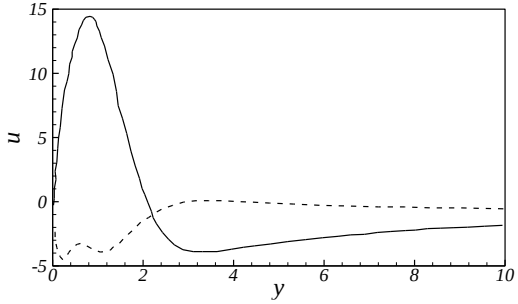




O-S方程的本征值和本征模态 $Re = 500$ $\alpha = 0.2$

$|c_i|$ 最小：最弱稳定模态





§6.1 湍流的产生

- **平面Couette流**
  线性稳定

- **圆管Hagen-Poiseuille流**
  线性稳定

§6.1 湍流的产生

- 粘性的双重作用

  根据**Rayleigh**定理

  - 射流、混合层、逆压梯度边界层等速度剖面存在拐点，有可能失稳；
  - 平面**Poiseuille**流、顺压梯度边界层等速度剖面没有拐点，是稳定的；但实际情况不是这样；

    粘性是引起不稳定性的关键因素
  - 粘性大，耗散强，是稳定的因素

§6.1 湍流的产生

- 临界层的概念
  - 线性扰动的相速度满足 $u_{\min} < c_r < u_{\max}$

    线性扰动的传播速度等于基本流某点的速度
  - 临界点：对于中性 $c_i = 0$ 情况，至少存在一点 y_c，使得 $u(y_c) = c$
  - 若 $D^2u(y_c) \neq 0$，**Rayleigh方程在临界点无解**

$$\left(D^2 - \alpha^2 - \beta^2\right)\hat{v} - \frac{D^2u}{u-c}\hat{v} = 0$$

在临界层附近必须考虑粘性

§6.1 湍流的产生

临界层的应用：风浪耦合问题

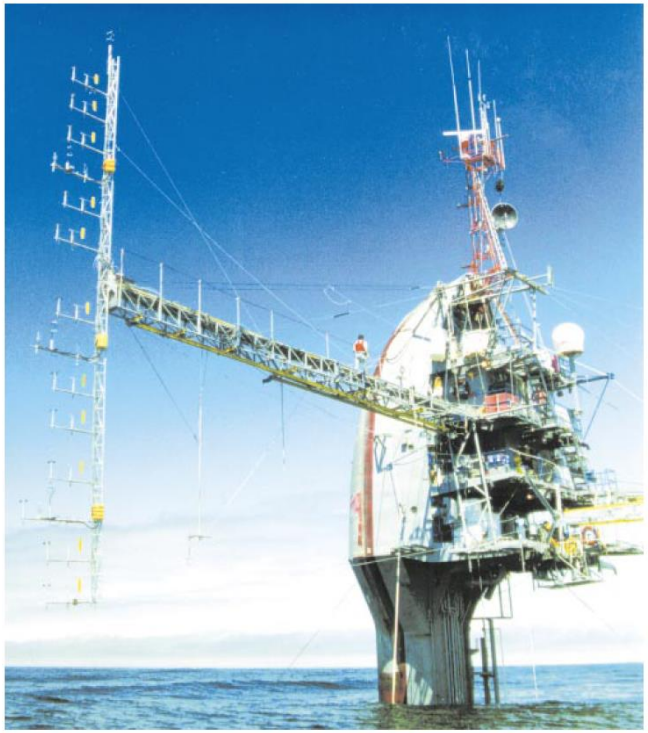


Figure 2 Floating Instrument Platform *FLIP* moored during the experiment. Four ultrasonic anemometers measured the wind velocity at different heights above the ocean surface and 12 cup anemometers and vanes registered the horizontal wind speed and direction. The sea surface elevation was measured beneath the mast.

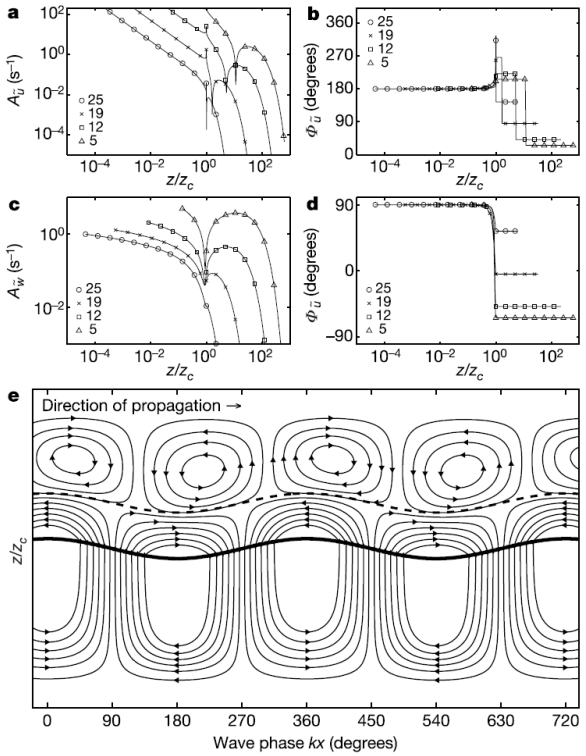


Figure 1 Numerical solutions for the Rayleigh equation (1). **a**, **b**, Amplitude $A_{\tilde{u}}$ (**a**) and phase $\Phi_{\tilde{u}}$ (**b**) of the horizontal wave-induced velocity fluctuations $\tilde{u}(z/z_c)$. **c**, **d**, Amplitude $A_{\tilde{w}}$ (**c**) and phase $\Phi_{\tilde{w}}$ (**d**) of the vertical wave-induced velocity fluctuations $\tilde{w}(z/z_c)$. Each curve corresponds to a value of the parameter c/u_*, as indicated in the symbol key. The flow streamlines (**e**) have been reconstructed from the solution ψ for $c/u_* = 5.75$. The solid line represents the air–water interface and the dashed line is the critical height. The whole eddy structure is propagating to the right following the wave. k is wavenumber.

Rayleigh方程数值解

波引起的垂向脉动跨临界层存在明显相位突变；c/U 越小，相位突变越明显。

Miles J W. On the generation of surface waves by shear flows. Journal of Fluid Mechanics, 1957, 3: 185-204.

Hristov T S, Miller S D, Friehe C A. Dynamical coupling of wind and ocean waves through wave-induced air flow. Nature, 2003, 422(6927):55-8.

§6.1 湍流的产生

临界层的应用：风浪耦合问题

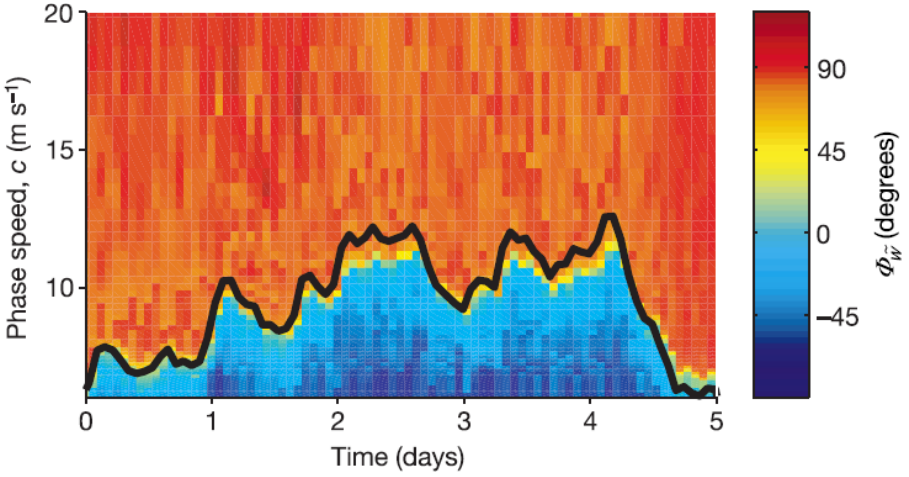


Figure 4 The measured phase of the vertical wave-induced velocity fluctuations $\Phi_{\tilde{w}}$ in degrees throughout a five-day period. The phase is referenced to the phase of the waves. The solid black line shows the mean wind speed $U(z_i)$ versus time, and also marks the wave mode $c = U(z_i)$, for which the instrument is at the mode's critical height. For modes faster than the mean wind the instrument is below their critical heights and for modes slower than the wind the instrument is above their critical heights. The line of the mean wind speed marks the persistent abrupt change in phase, as predicted by the solutions of the Rayleigh equation (1), and in agreement with Figs 1d and 3d. Data are from the anemometer at $z_i = 3.5$ m.

监测结果与理论值的比较

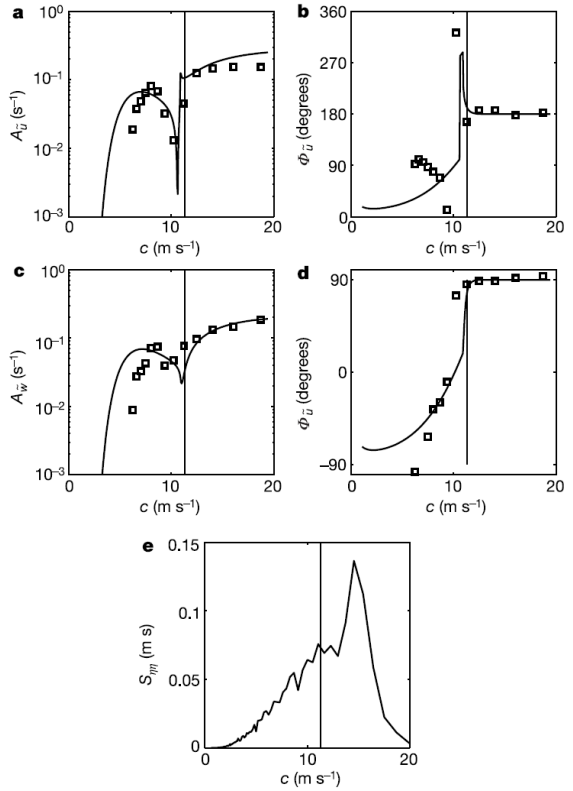
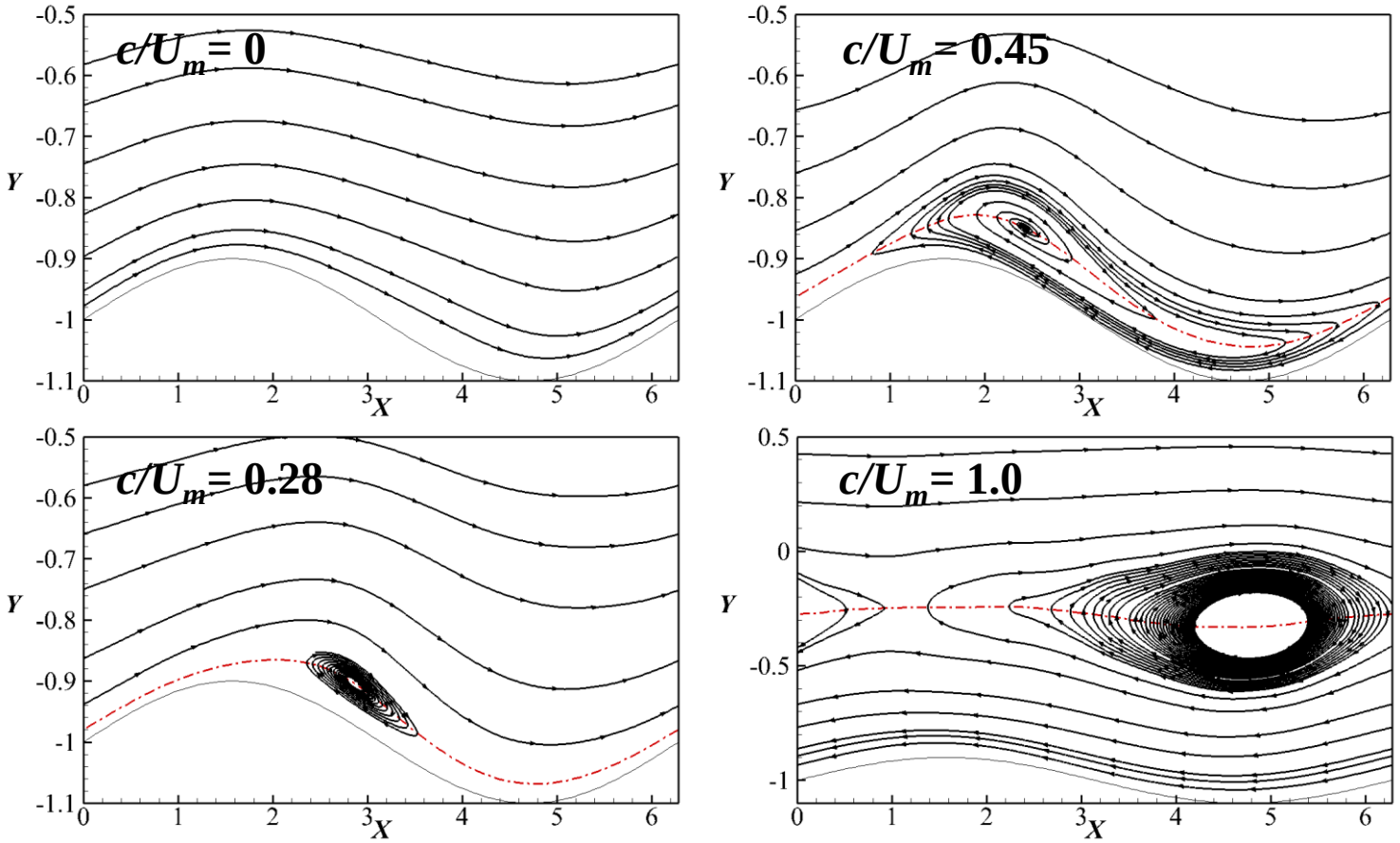


Figure 3 Theoretical and experimental transfer functions for the wave-coherent velocities $T_{\tilde{u}}(c) = A_{\tilde{u}}e^{i\Phi_{\tilde{u}}}$ and $T_{\tilde{w}}(c) = A_{\tilde{w}}e^{i\Phi_{\tilde{w}}}$ and a surface elevation power spectrum. **a**, **b**, Amplitude $A_{\tilde{u}}$ (**a**) and phase shift $\Phi_{\tilde{u}}$ (**b**) for the along-wind wave-induced velocity fluctuations $\tilde{u}$. **c**, **d**, Amplitude $A_{\tilde{w}}$ (**c**) and phase shift $\Phi_{\tilde{w}}$ (**d**) for the vertical wave-induced velocity fluctuations $\tilde{w}$; (**e**), Surface wave power spectrum $S_{\eta\eta}$ versus the wave phase speed c. The amplitudes in **a** and **c** are normalized by the surface wave amplitude at that phase speed; the phases in **b** and **d** are referenced to the phase of the waves. The solid lines are numerical results from equation (1); the squares are experimental estimates from time series of wind velocity ($z_i = 8.7$ m) and wave elevation collected over 90 minutes. The vertical line marks the mean wind speed measured over the same time period, that is, it shows the wave mode for which the instrument is at the mode's critical height.

§6.1 湍流的产生

DNS结果：
Re_τ=300

相位平均的流线（随波参考系下）

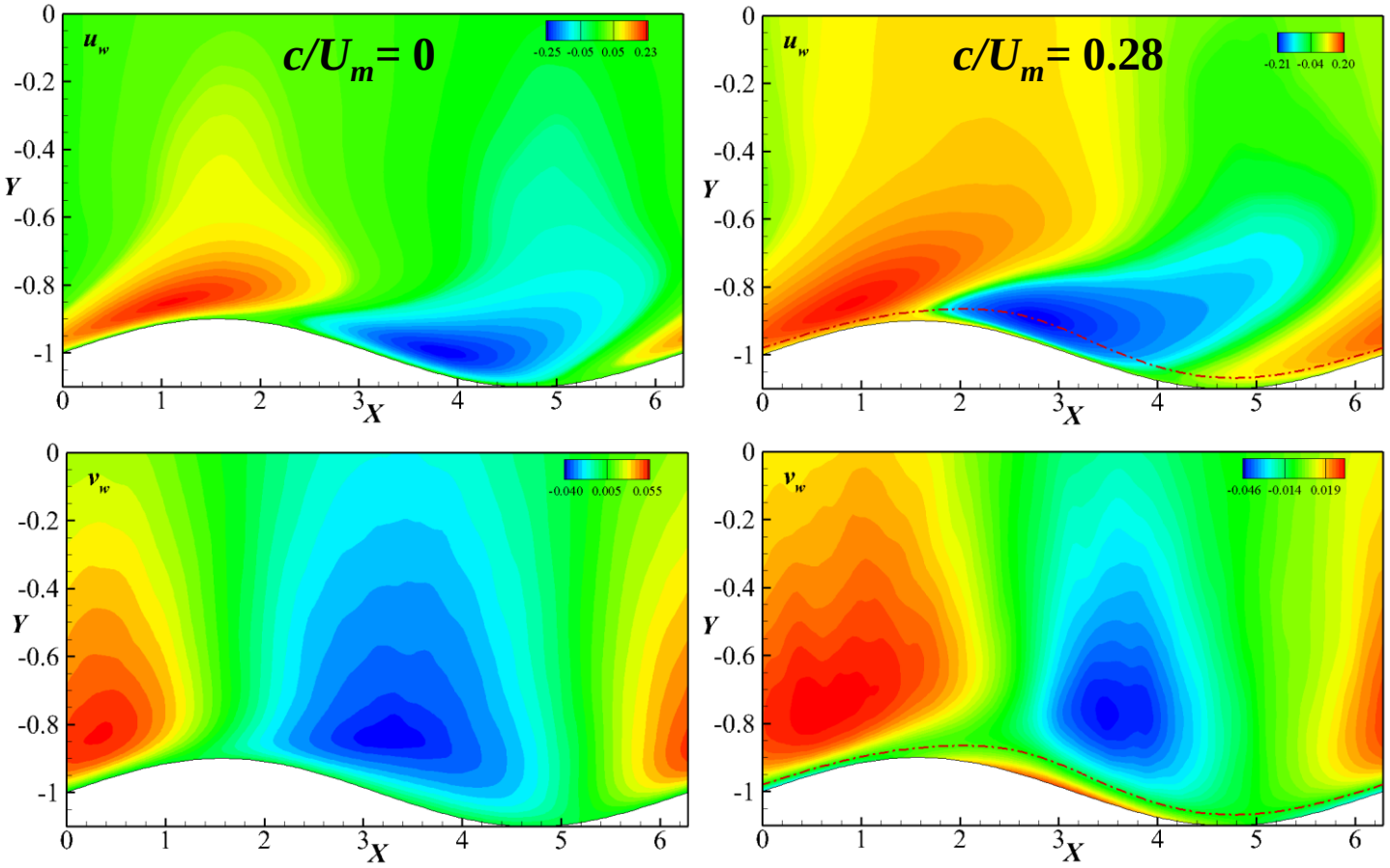


对于静止波形壁，近壁流动在迎风面加速，背风面减速；行波时，临界层出现，临界层沿流向不对称，迎风面较薄，背风面较厚，猫眼涡的中心在临界层高度位置，随着波龄的增加，临界层上移，变得更平坦，当$c/U_m > 1.0$时，临界层在计算域外。

§6.1 湍流的产生

DNS结果：
Re_τ=300

波引起的流向与法向速度脉动



跨临界层：
波引起的流向速度脉动没有明显的相位突变
波引起的垂向速度脉动存在明显的相位突变

§6.1 湍流的产生

讨论：粘性对流动稳定性所起的作用，加强稳定还是破坏稳定？

历史上一直错误地认为：粘性对流动稳定总是有益的，其实不然！

为从理论上分析粘性会导致流动不稳定的可能，Heisenberg (1924)从O-S方程出发，在大雷诺数下把解进行渐近展开，发现流动出现了不稳定，但他没有能求出失稳的临界雷诺数。

由于对Heisenberg使用的渐近展开法的可靠性有较大疑问，又不能解释雷诺数趋于无穷时流动趋于中性稳定的原因，很长一段历史时期该佯谬成为流体力学界争议的一个热点问题。

C.C.Lin(1945)用令人信服的数学理论对渐近展开法做了全面论证，阐述了这种方法的有效性，在提高计算精度的基础上得到了临界雷诺数。这种方法现在被称为摄动法，并有广泛的应用。

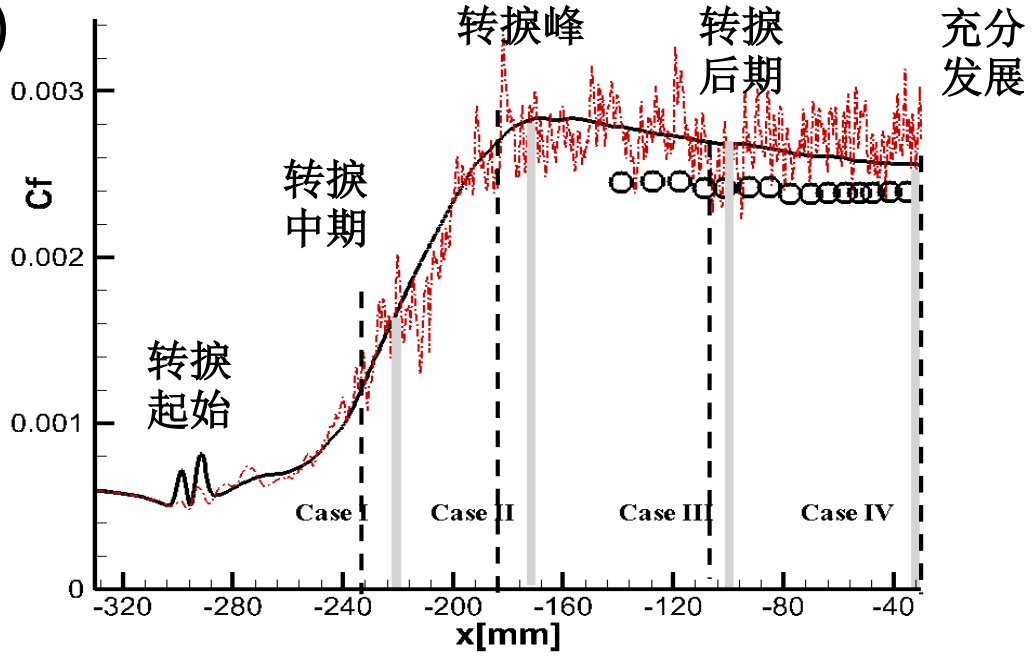
Thomas(1953)用电子计算机得到了相近的结果。

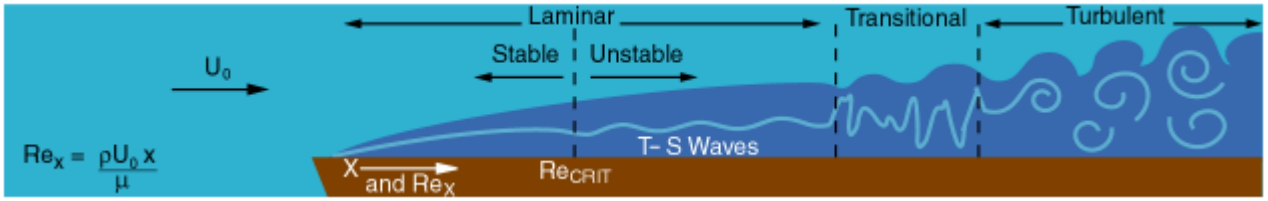
结论：粘性对扰动的发展具有双重作用，一方面，它会消耗扰动的能量，抑制扰动的增长；在一定的情况下，它也会通过粘性将能量从主流传给扰动，使流动不稳定。

§6.1 湍流的产生

六、二维平板层流边界层的转捩过程

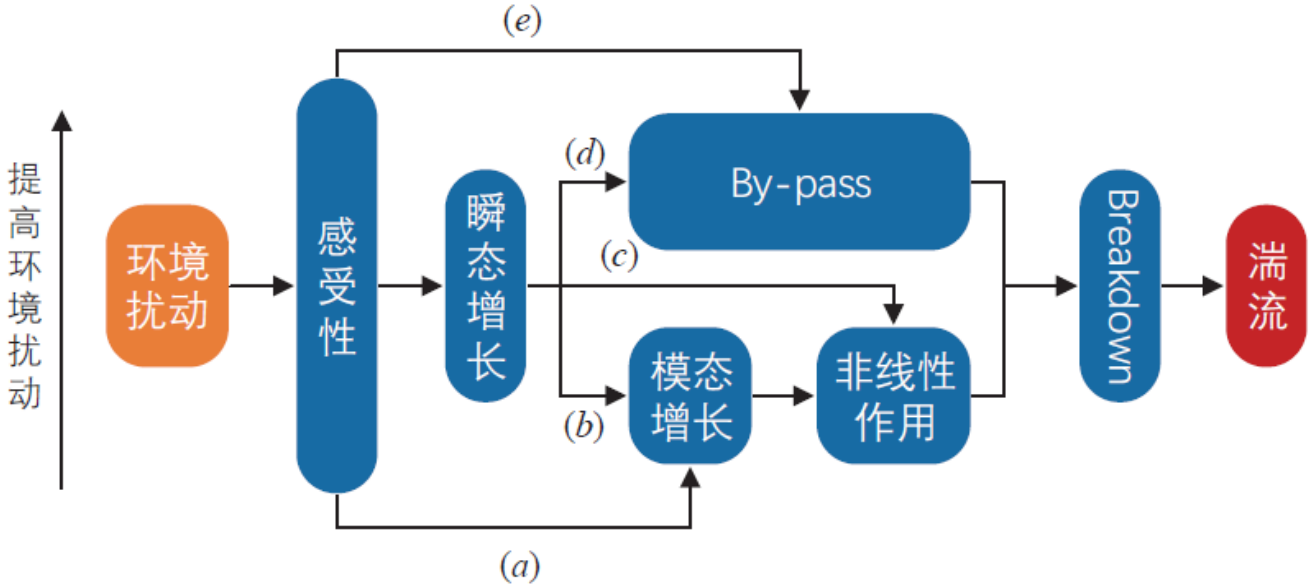
- 扰动接收(receptivity)
- 线性增长
- 非线性相互作用
- 二次失稳
- 破碎 (breakdown)





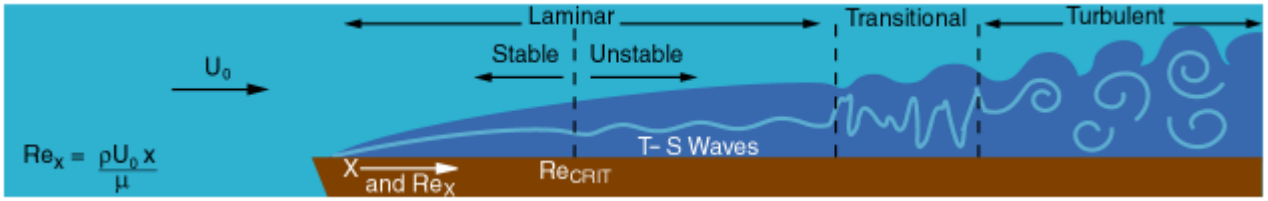
§6.1 湍流的产生

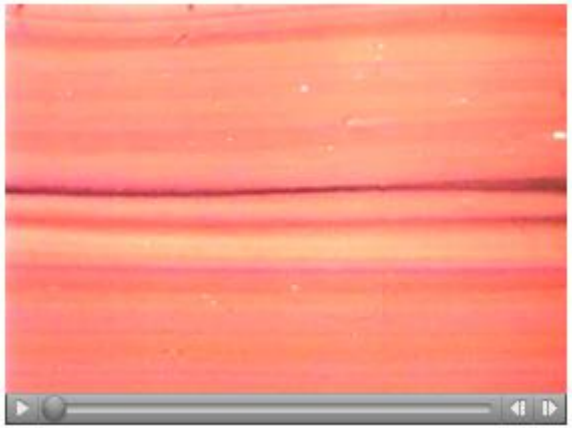
- **正规转捩**
  - 扰动较小时，边界层经历线性失稳、扰动进一步放大、不同扰动模态之间非线性相互作用，最后转变为湍流
- **旁路转捩**
  - 扰动较大时，边界层的层流流动跳过线性阶段，直接经过非线性演化而转变为湍流



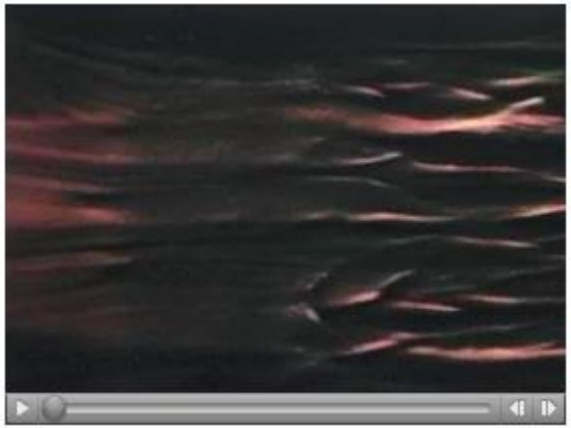
§6.1 湍流的产生

边界层的正规转捩





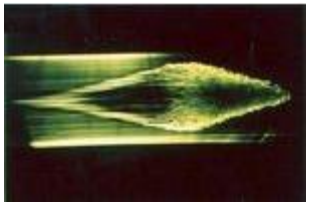
层流



T-S波

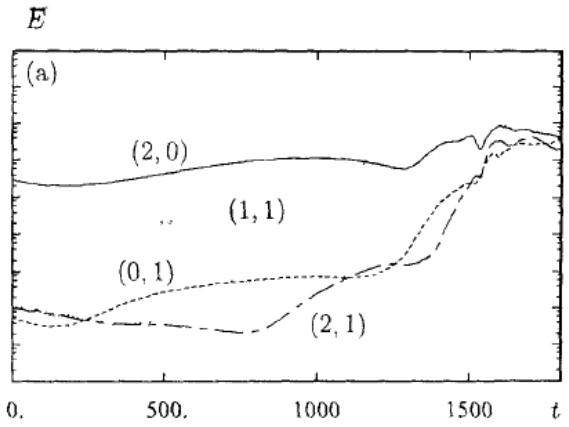


湍斑

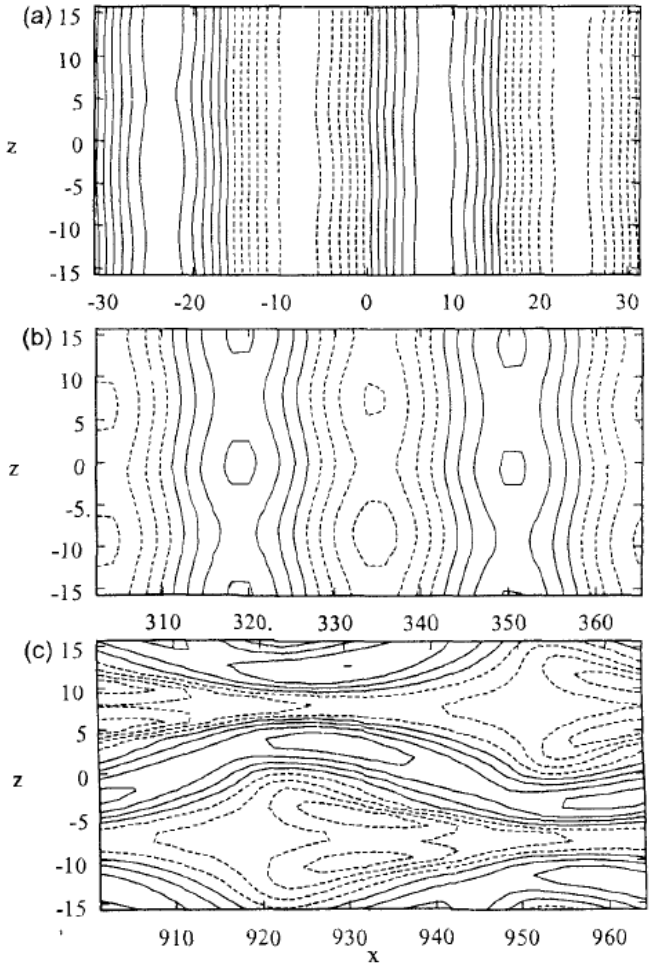


§6.1 湍流的产生

转捩机制1：T-S波的二次失稳



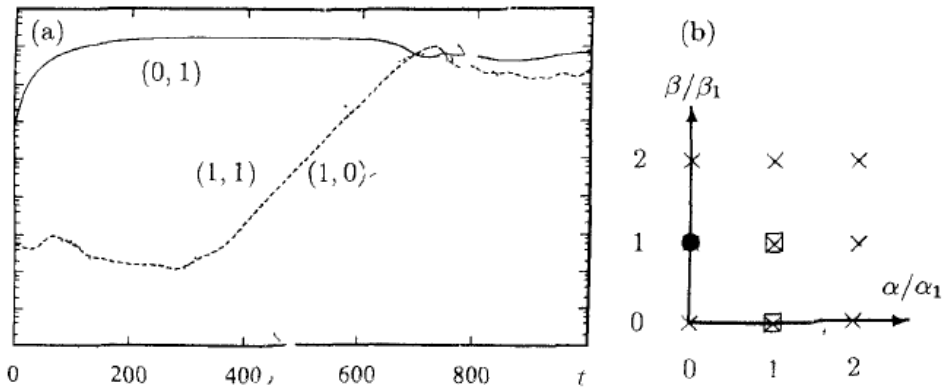
各波数上的能量变化



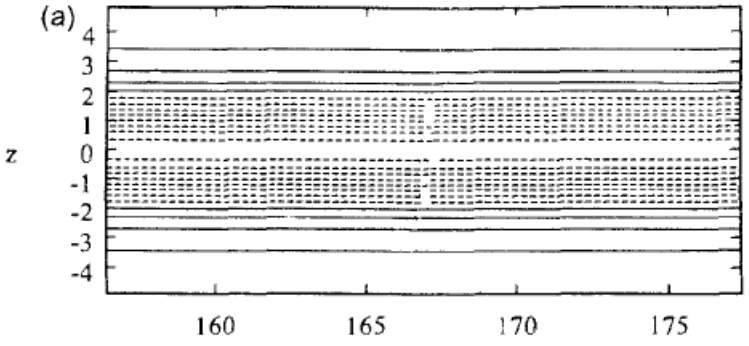
§6.1 湍流的产生

转捩机制2：条带破碎

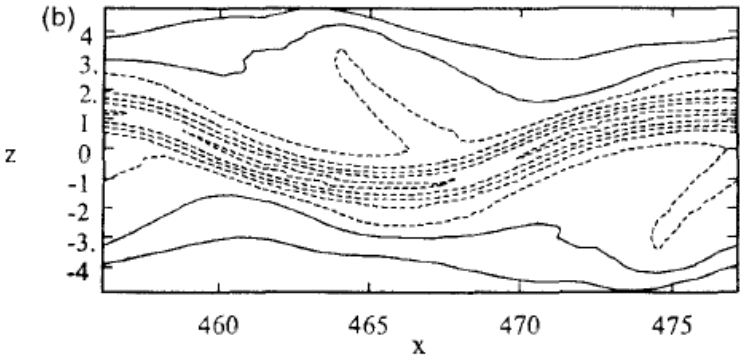
各波数上的能量变化



t=250



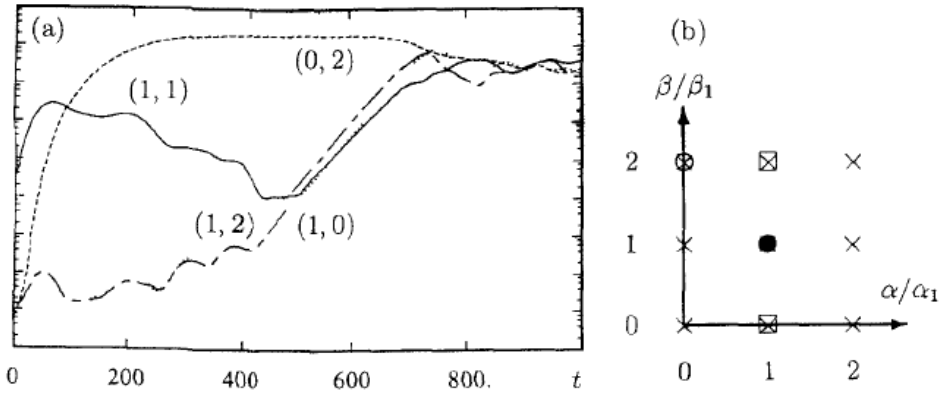
t=700



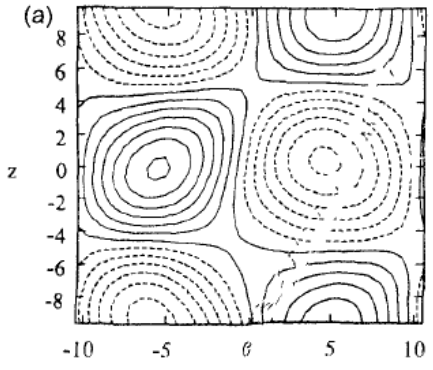
§6.1 湍流的产生

转捩机制3：偏斜转捩

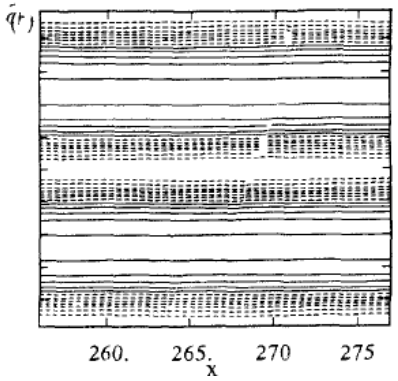
E 各波数上的能量变化



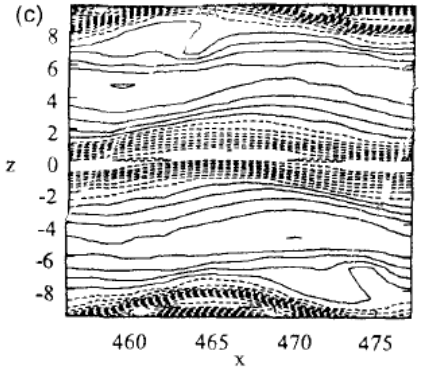
t=0



t=400



t=700

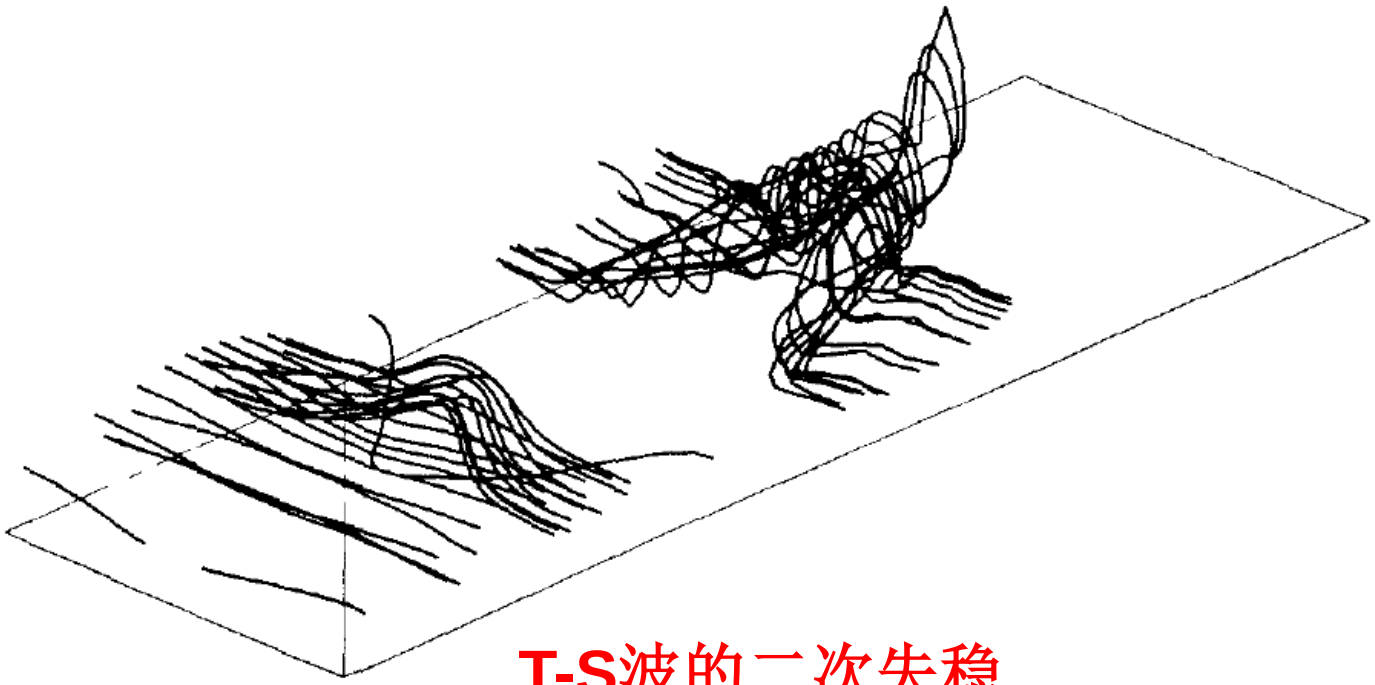


§6.1 湍流的产生

转捩过程的影响因素

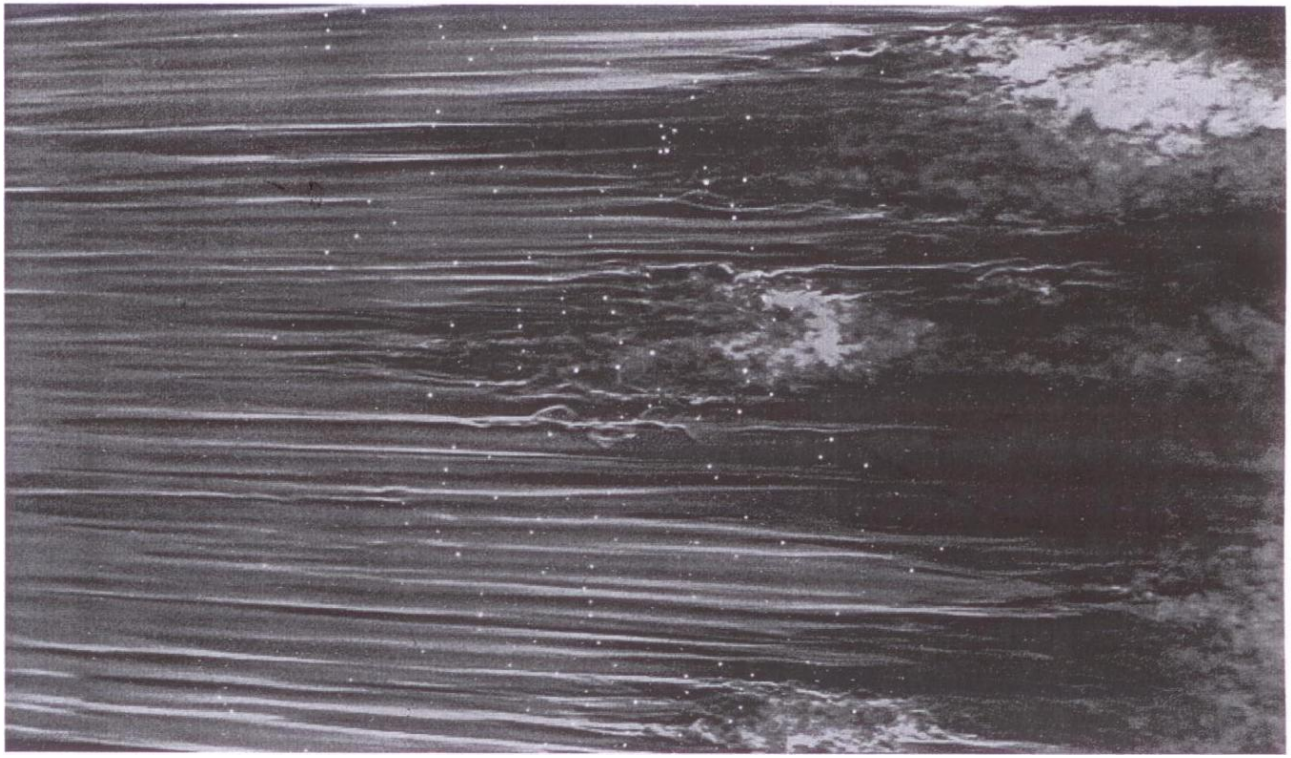
初始扰动谱、扰动幅值分布、二次失稳的阈值

不同的条件下将由不同的转捩机制主导



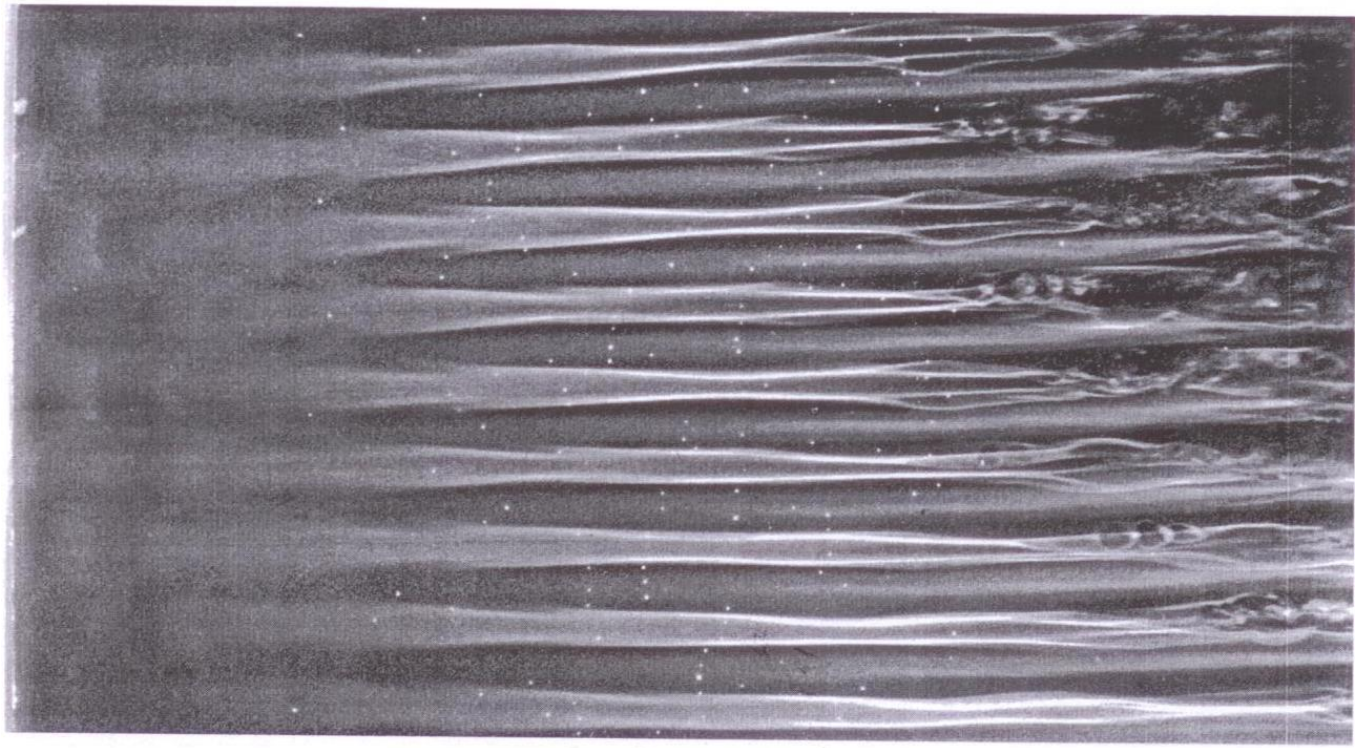
T-S波的二次失稳

§6.1 湍流的产生



条带结构的破碎

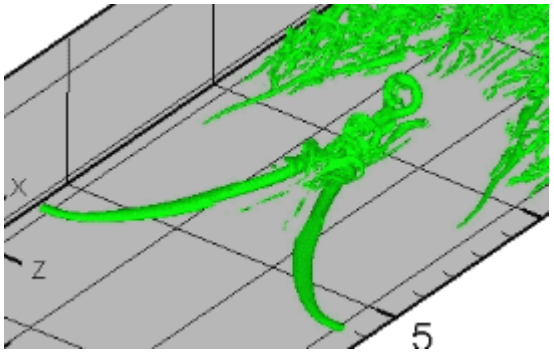
§6.1 湍流的产生

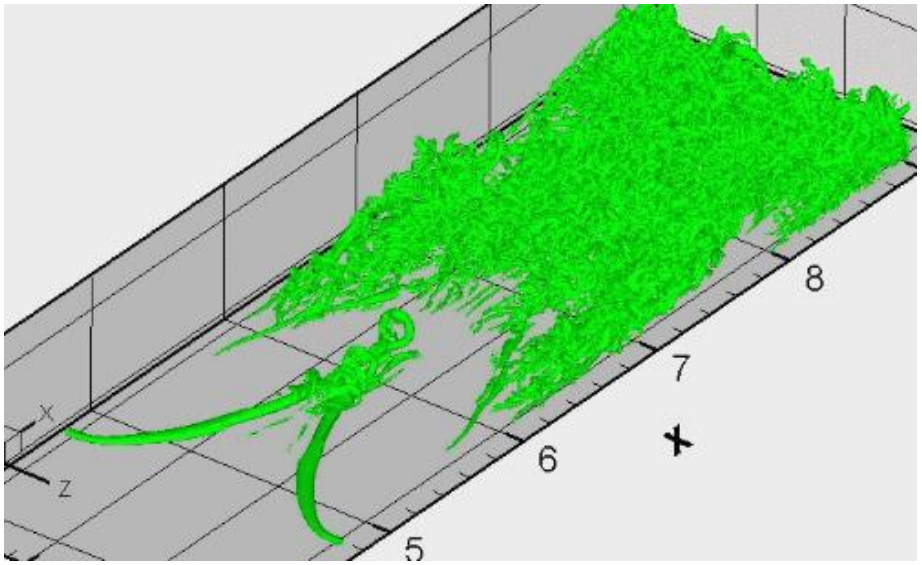


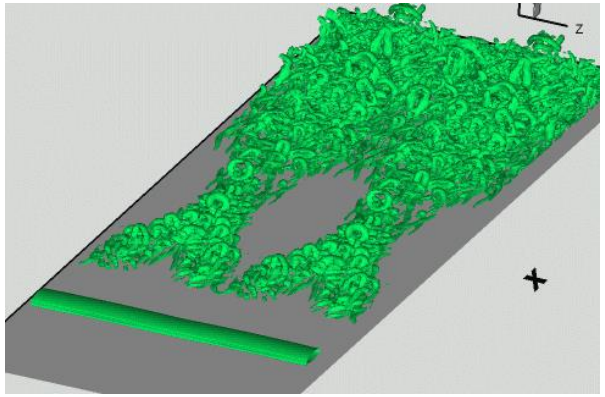
偏斜转捩

§6.1 湍流的产生

动画演示： 平板边界层拟序结构的形成及演化





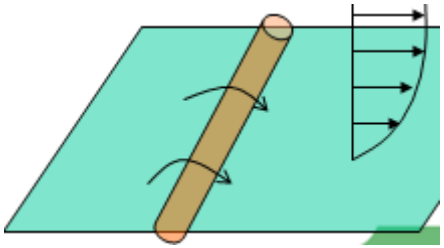


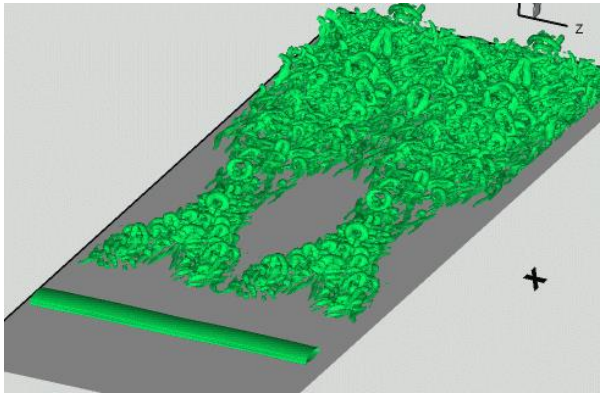
流场中涡结构的瞬时演化图（Q=10的等值面）

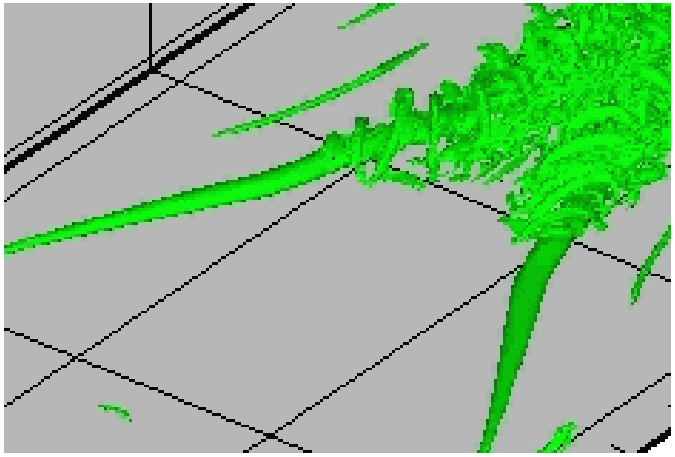
参考：傅德薰、马延文、李新亮、王强， 可压缩湍流直接数值模拟， 科学出版社

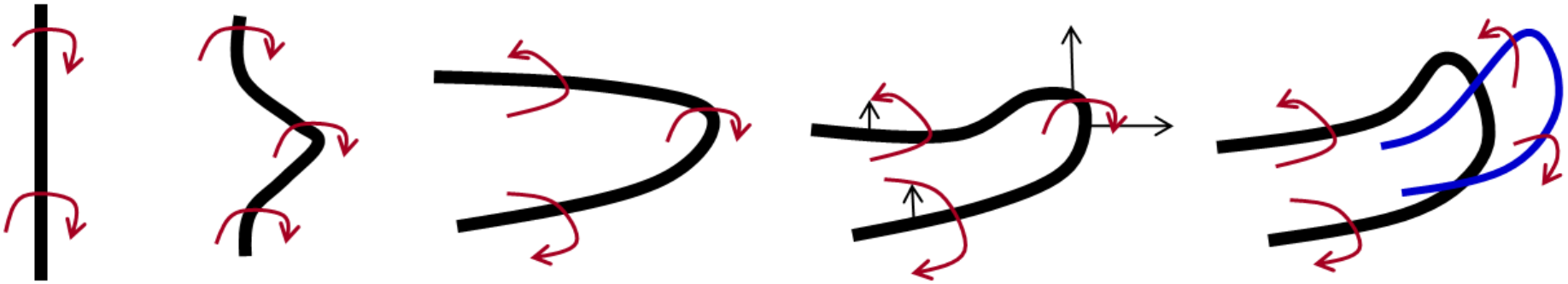
§6.1 湍流的产生

发卡涡形成及发展的涡动力学机制







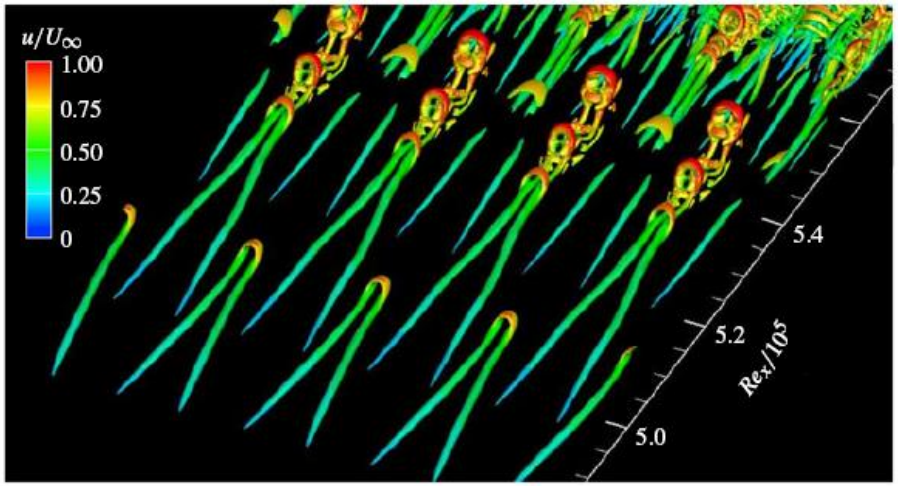


二维涡丝　三维变形　　自诱导头部抬起，拉伸　　诱导出新的涡

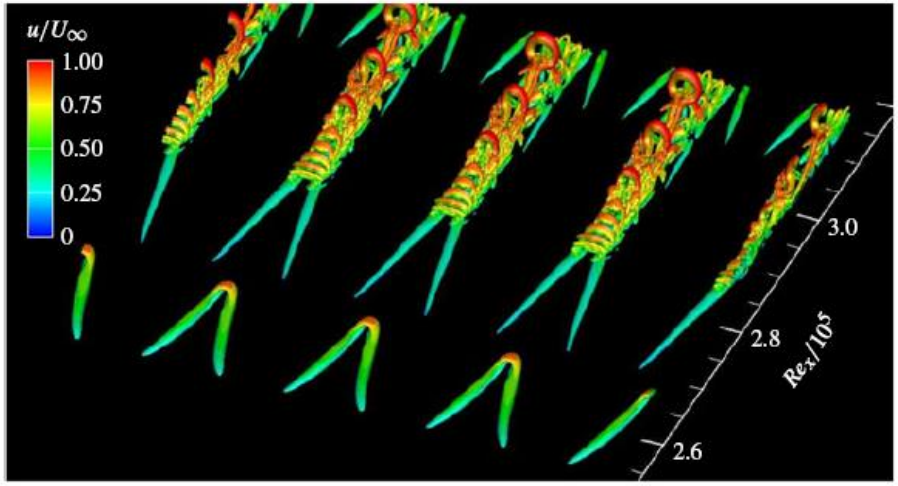
参考：傅德薰、马延文、李新亮、王强， 可压缩湍流直接数值模拟， 科学出版社

§6.1 湍流的产生

直接数值模拟结果



H-type transition



K-type transition

	L_x	L_y	L_z	N_x	N_y	N_z	Δx^+	Δy^+_{min}	Δz^+	$Re_{\theta_{max}}$	λ_0
H-type	10.6	1.00	0.6	4096	550	512	10.3	0.41	5.4	1250	0.15
K-type	9.6	0.92	0.6	4096	550	512	9.7	0.39	5.5	1410	0.15

TABLE 1. Characteristics of the H- and K-type simulations.

参考： **Sayadi, Hamman, Moin (2009)**

§6.1 湍流的产生

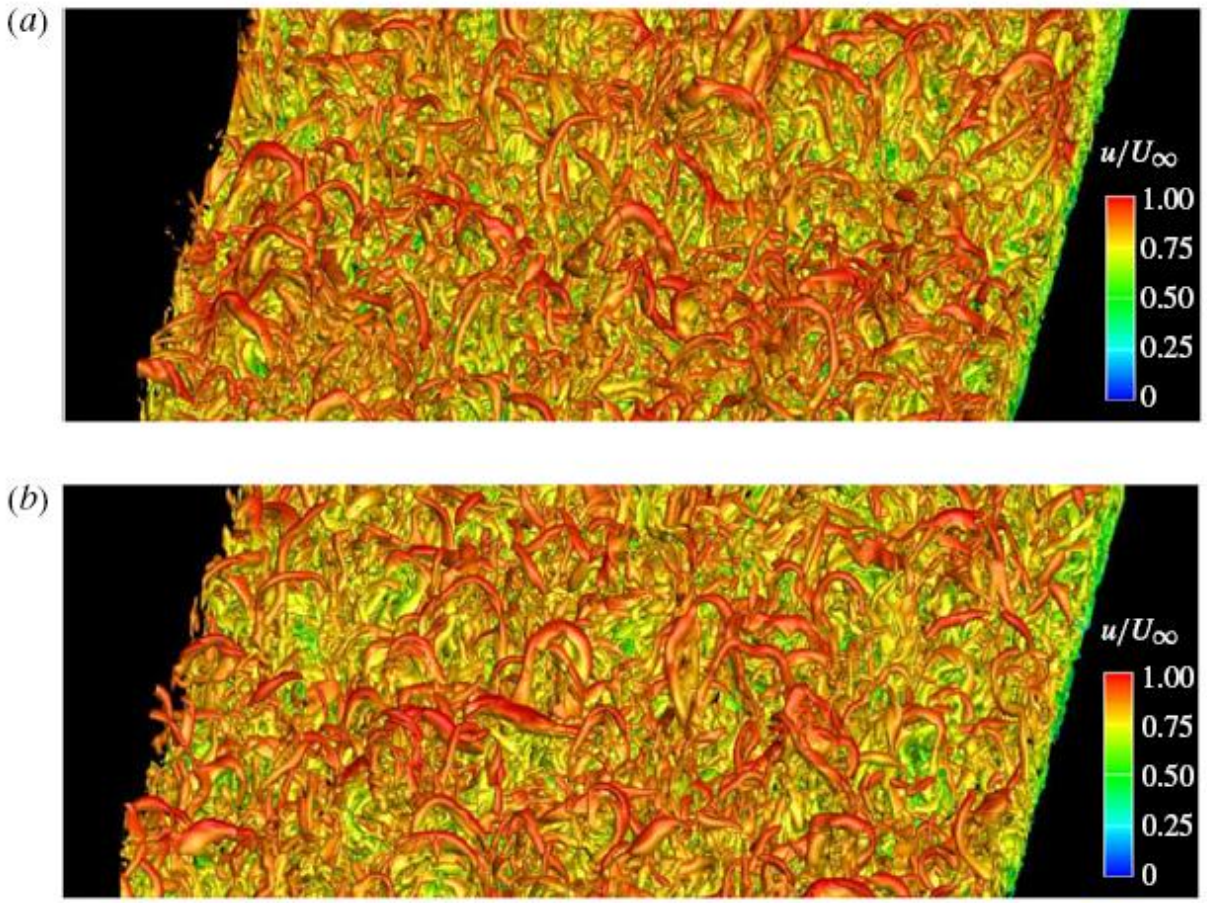
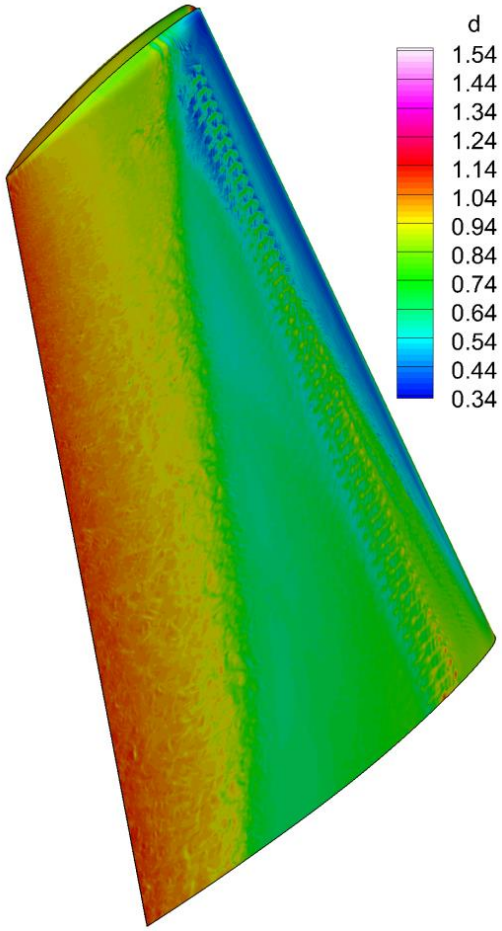


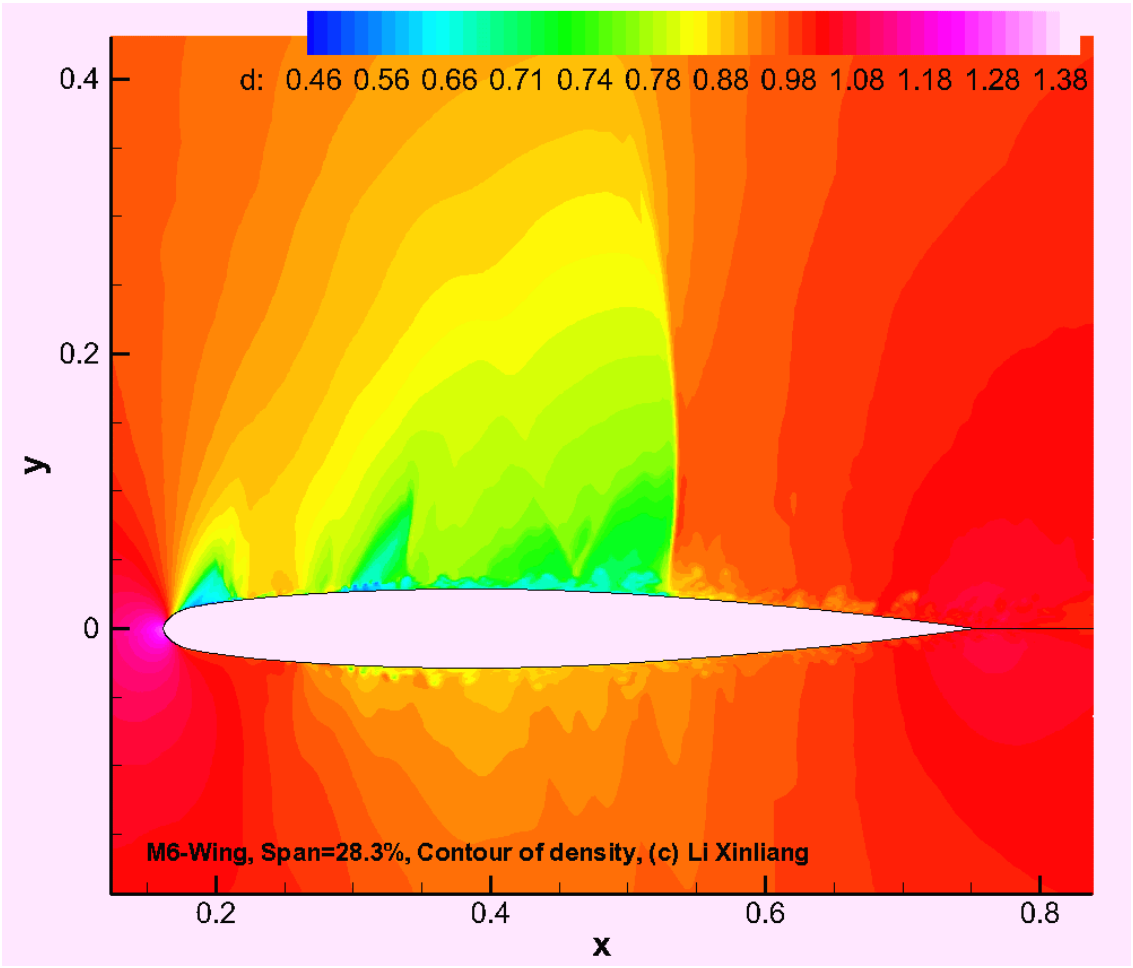
FIGURE 12. Instantaneous isosurfaces of the second invariant of the velocity gradient tensor, Q, coloured by streamwise velocity, in the fully turbulent region near $Re_\theta = 1200$: (*a*) H-type, $Re_x = 8.7 \times 10^5$; (*b*) K-type, $Re_x = 6.8 \times 10^5$.

§6.1 湍流的产生

ONEAR-M6 翼， Ma=0.8395，Re=1.17×10^7（李新亮提供）



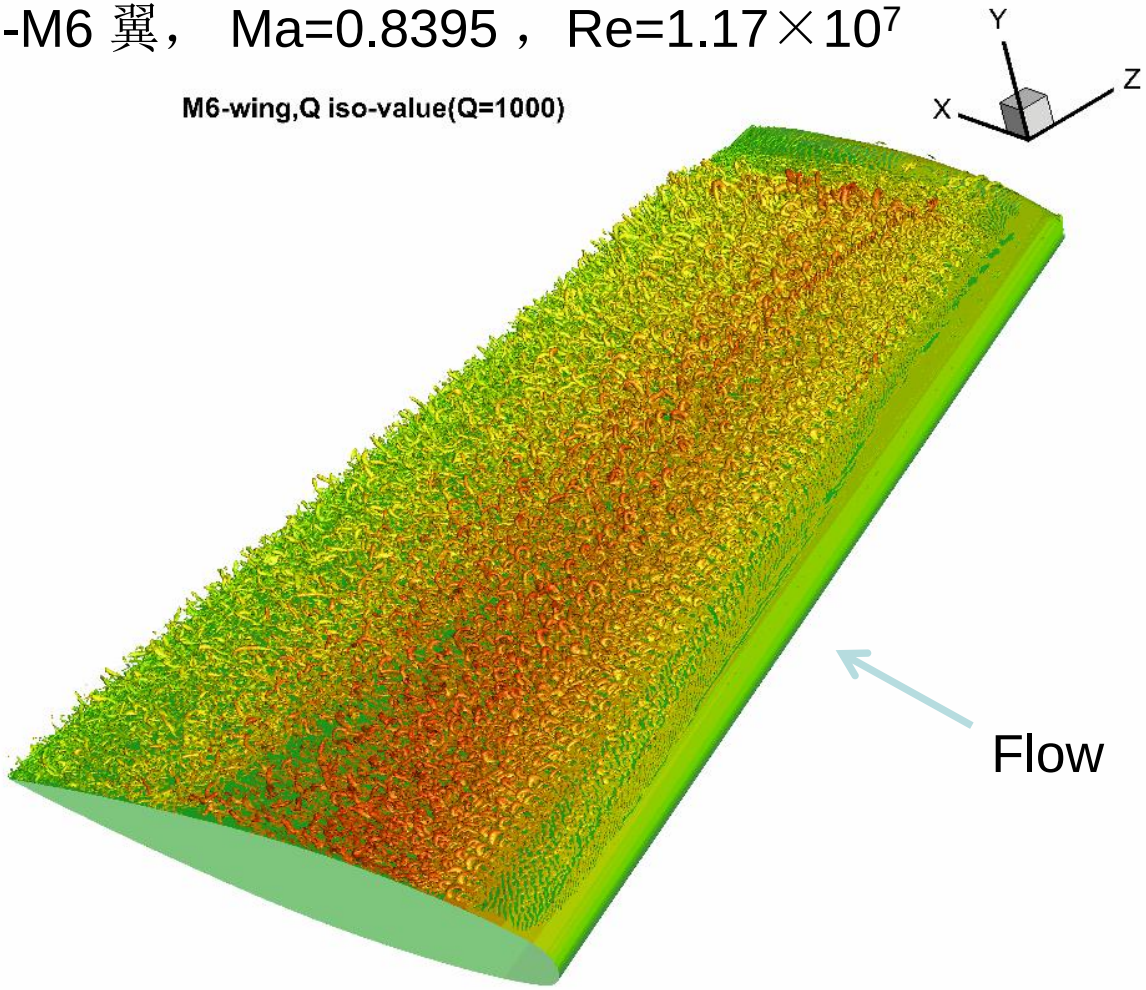
翼面瞬时密度分布



翼面及 **z=28.3%** 截面上瞬时密度分布

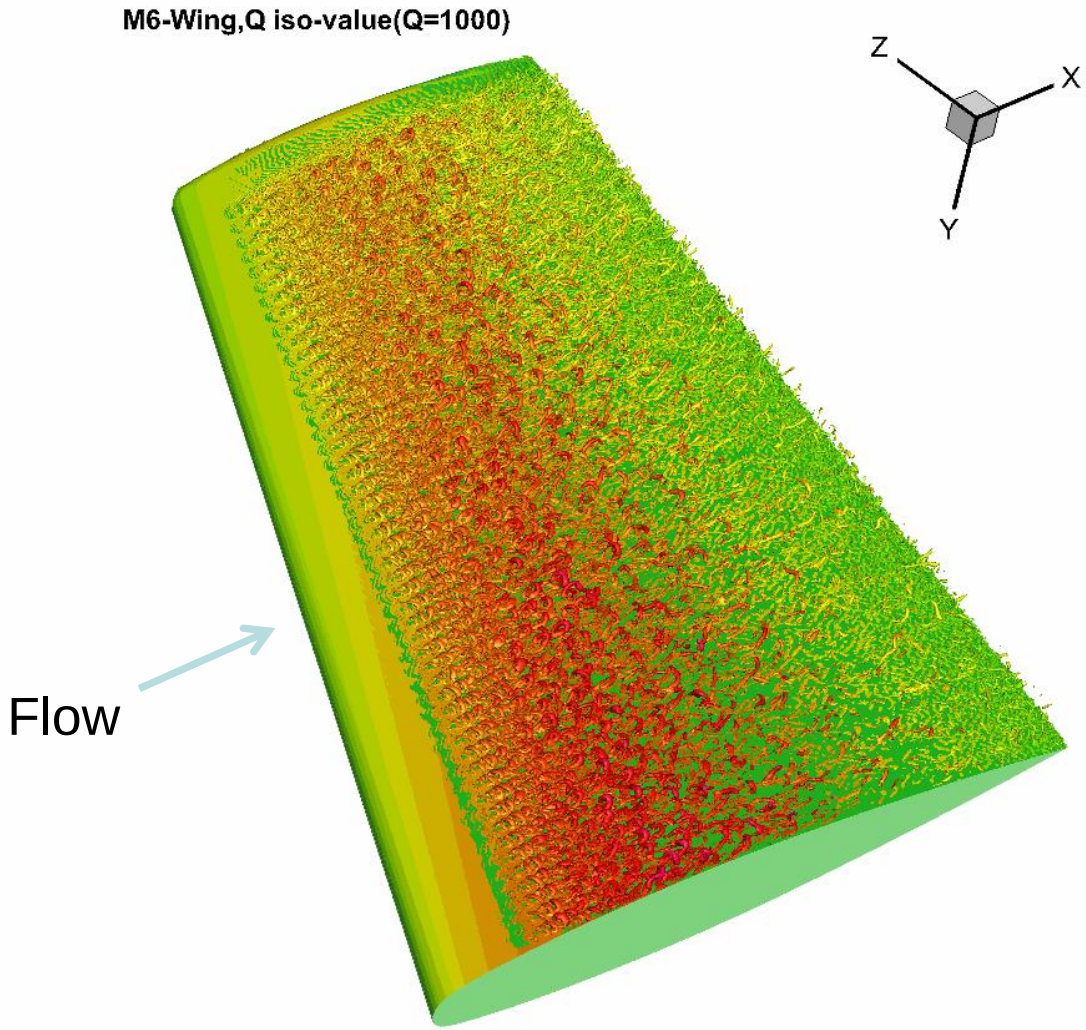
§6.1 湍流的产生

ONEAR-M6 翼， Ma=0.8395，Re=1.17×10^7



ONEAR-M6 三维翼 上表面的涡结构

§6.1 湍流的产生



ONEAR-M6 三维翼 下表面的涡结构

§6.1 湍流的产生

- 所有层流流动在速度足够高的情况下都会失稳转捩为湍流；
  - 雷诺数、泰勒数
- 稳定性问题可以采用基本流叠加扰动的方式进行研究；
- 转捩过程的研究至今还是个难题；
- 其它因素的影响：
  - 压力梯度，壁面吹吸，柔性壁面，壁面温度，本构关系
- 其它稳定性问题：
  - 边界层流动、热边界层、可压缩流动、非定常流动、曲面边界层